

SOCIAL MEDIA LITERATURE REVIEW

TOWARD A SECOND BRAIN, READ AGAINST SOCIAL MEDIA’S 2024-2026 AGENT FOLKLORE

- **Date:** 2026-08-18
- **Status:** literature review / design instrument (not a spec, not an implementation plan)
- **Length:** ~191 000 characters (target band ≥ ~100 000)
- **Corpus:** curated Social Media posts
- **Companion:** `AI_agents_literature_review.md` (eleven books + Karpathy LLM-wiki cluster, same date, same stance)
- **End goal:** the same as the book review — a durable approach to a personal “second brain” whose agents can *act* (task-oriented) and *know* (knowledge-oriented) without collapsing those two orientations into a chatbot, a vector index, a prompt pack, or a second control plane
- **Related (ASC, 2026-08):** `Projet Complexe 2026 Revival (v1 and v2)`; `Four Layers`; `Cognitive Institutions`; `IEML-as-compass`; `Reasoning without probabilistic inference`; `Meadows leverage points`; `Reverse prompting / Cognitive Load Ratio`; `Agents of Redirection`; `Le Moigne`; `Lefèvre`; `What is Ecological Redirection`
- **Related (Projet Complexe, 2026-08):** `14-proposed-architecture.md`; `17-local-dev-stack-architecture.md`; `17-ui-design-ideas.md`; `18-graph-rag-wikipedia-db-pedia-ieml.md`

This document is a **working instrument**. It exists so that later pivots (`index`, `extract`, `relate`, `research`, `run-agent`, `inspect-agent`, `stop-agent`, `publish`) can be chosen with a map of what the 2024–2026 *Social Media AI genre* actually taught, instead of treating saved carousels as a second bookshelf. It is **not** a substitute for the posts, the papers they compress, or the book review. It is a **reading of a social-media corpus from the same architectural stance** that already framed the books. Where a carousel repeats a book, this review records the *folk version* — what actually circulates — and asks whether that folk version helps or harms *Projet Complexe*.

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0. HOW THIS REVIEW WAS PRODUCED, AND HOW TO USE IT

0.1 Method

The source folder is an Social Media **save archive**: `saved/YYYY/MM/DD/<account>/<shortcode>.txt` plus images (`.jpg` / `.webp` / `.png`), videos (`.mp4`), and metadata (`.json.xz`). About 3746 caption files exist across 2022–2026. About 1822 posts include a video. The first pass skipped every post that had an `.mp4`. A later pass **reads the caption** of those posts with the same AI / *Projet Complexe* filter as the stills. A third pass ran **speech-to-text on ten clips** whose captions were teasers (Shift-on-radio, BFM, Abel Quentin, Meyronnis, two Blast labour shows, Sarandi, two Ninon explainers, Carbonell book rec). It does **not** OCR poster frames or treat moving image as a knowledge plane. The other ~1800 videos stay caption-only.

The review **paraphrases**. Dumping captions would be both a copyright problem and a design problem: a second brain that stores other people’s carousels as its own thoughts is the same failure mode the 2026 notes already named for Wikipedia dumps (a library, not an import). Terms of art are kept (`ReAct`, `MCP`, `RAGAS`, `CAG`, `MAG`, `Graphify`, `Darwin Gödel Machine`, `attention sinks`, `GSM-Symbolic`). Arguments are restated in the vocabulary of *Projet Complexe*: Task, Implementation, Requirement, Fallback, Claim, Link, Factor, KnowledgeGap, pivot, hook, event, page, LOD. Each major claim is asked the same four questions used in the book review:

1. **What does the post actually claim?** (not the hook, not the “comment PROMPTS” funnel)
2. **How do they implement it?** (framework, protocol, eval, hardware assumption)
3. **Where does it sit on the August 2026 stack?** (ASC / *Projet Complexe* ASC / *Projet Complexe* / Compose engine / UI)
4. **Steal, adapt, or refuse?** for a local, multi-provider, filesystem-first second brain

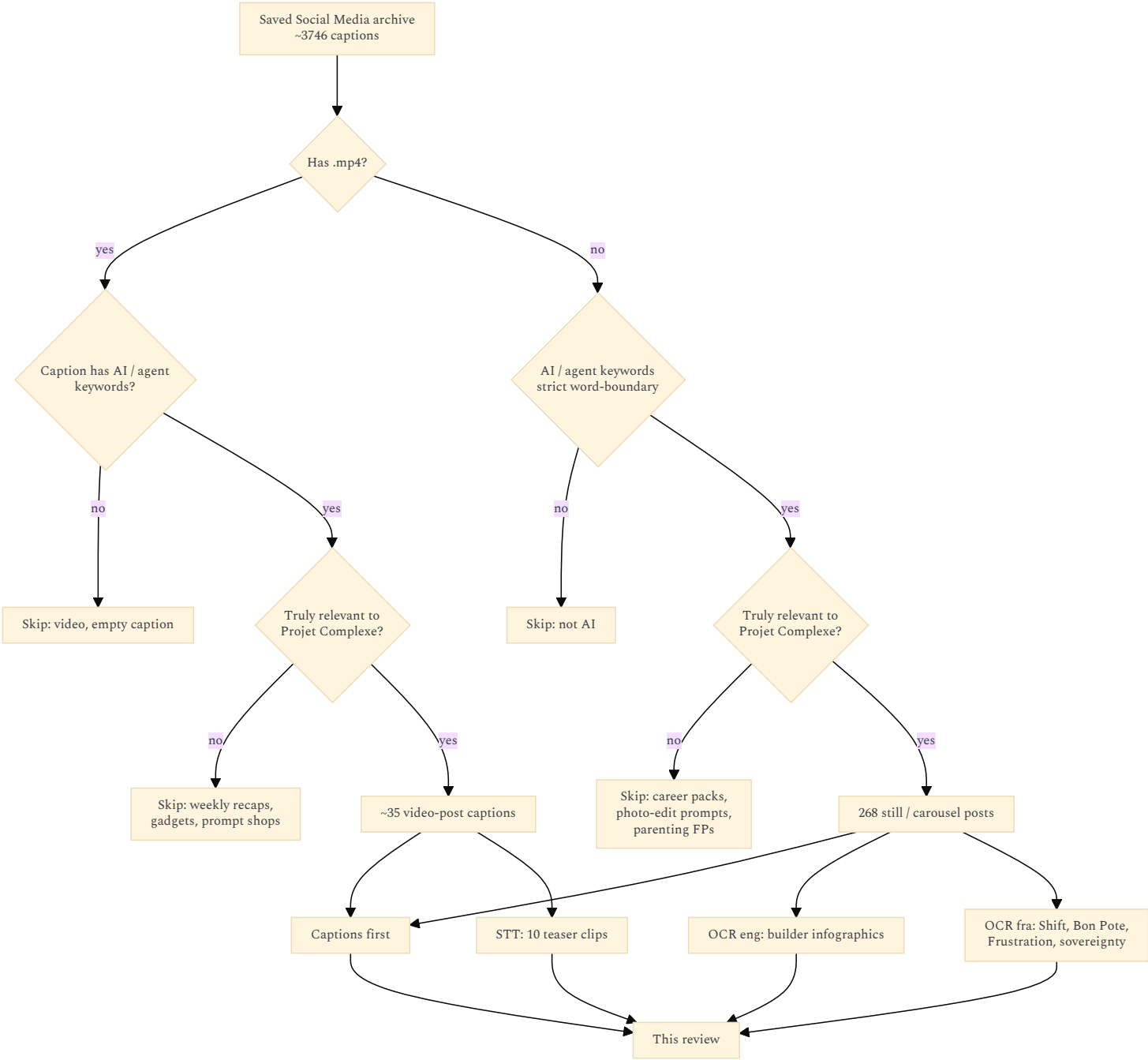
0.2 Filter

A first keyword pass was a false positive (a verbose regex treated `#(? :ai|... as a comment, so every caption matched). The working filter is strict: word-boundary matches on LLM / GPT / Claude / Gemini / agentic / RAG / MCP / transformer / and a short list of known AI accounts, plus a manual skip of remaining false friends (parenting, iPhone settings, “old internet gems”, career-pivot prompt packs that mention Claude only as a job coach, Musk productivity emails, confessional-booth novelties). Having an .mp4 is not a skip. The skip is an empty or off-topic caption.`

After that cut: **268 still / carousel posts, 64 accounts, 2597 images**, almost all from 2025–2026 (217 in 2026, 41 in 2025, 9 in 2024, 1 in 2023). 238 are carousels of four or more slides. Caption text of the filtered stills is about 262000 characters. That number is misleading: on Social Media, the argument often lives **on the slides**, and the caption is a teaser plus a hashtag block plus a paid-PDF funnel.

Video-post captions (this pass). Same keyword/account filter on the ~1822 posts that have an `.mp4`: **107 matches**. Most are `evolving.ai` weekly recaps, product demos, or prompt shops already covered by stills — those are not spent pages on. ~**35 captions** add an argument the stills did not (local-offline stacks, CAG hot/cold, agent-root warnings, labour “observed exposure,” Brazilian campus siting, Shift-on-radio). Those are folded into Parts II–VIII below. Caption text of the 107 is about 102000 characters; the synthesis uses the ~35, not the recap bulk. **Ten of those teasers** were then transcribed (\$0.3b); the rest stay caption-only.

Top still accounts that survived the filter (by post count): `evolving.ai` (72), `datasciencebrain` (30), `thewizeai` (22), `theartificialintelligens` (12), `airesearches` (10), `rossfledderjohn` (9), then a long tail of prompt shops, open-source roundups, and a small French critical cluster (`mediapart`, `mondediplo`, `contretemps_revue`, `theshiftproject_`, `bonpote`, `lundi.am`, `frustrationmag`, `viepubliquefr`, `cairn.info`). Video captions add Blast, Bolchegeek, France Inter, BFM, `ninon.ia_officiel`, and a Brazilian data-center thread (`matheuspggomes`).



0.3 OCR

System `tesseract-ocr` is not installed as a Debian package. OCR used the Frog / GNOME Builder staging binary (`tesseract 5.5.0`). Two language passes:

- `eng` — English educational carousels (`datasciencebrain`, `techwith.ram`, `thewizeai`, `vyzual.ai`). Recoverable at the level of arguments and tables. Code on slides is truncated and mis-cased.
- `fra` (`fra.traineddata` from `tessdata_best`, `--psm 6`) — French infographic carousels whose captions were short relative to slide count: Shift Project series 1 and 2, Bon Pote, Frustration, Vie publique, Contretemps (Frank, Tanuro), Lundi matin (Griziotti), `cairn.info`. Re-OCR of Shift-1 / Bon Pote with `fra` fixed the earlier English-model garble (`Historiguement` → `Historiquement`, `d'usage` readable). Charts and overlaid numbers are still lossy (growth-rate percentages, France 2035 shares, Google’s potable-water fraction, GPU lifetime in years). Arguments and named scenarios survive.

OCR was **not** run on all 2597 images. Skipped on purpose: `evolving.ai` news recaps (captions already long), prompt-shop posters, Mediapart photographs, videos’ poster frames (the video pass uses **captions**, not frames). One wrong-target: a Contretemps anti-imperialism carousel (Rosato) sat next to Joshua

Frank in the save folder and was OCR'd; it is not used below.
OCR is treated as **evidence of what the slide claimed**, not as a verbatim extract to paste into the second brain. Slide text is paraphrased the same way captions are.

0.3b Speech-to-text (ten clips)

Entry point: `asc/extensions/transcription/transcribe/file.sh` (ffmpeg → 16 kHz mono wav → faster-whisper **medium**, auto language). Host: CPU (`CUDA_VISIBLE_DEVICES` empty; broken cuDNN on this machine). Working copies under `/tmp/ig-synth/stt/` so the Social Media save folder is not filled with `.wav`.
Ten files, ~17 minutes of audio, French except Sarandi (Portuguese). Whisper is treated like OCR: **evidence of what was said**, paraphrased, names corrected where the model garbled them (ChatGPT, AlphaFold, Moltbot, Sarandi, Carbonell, taylorisme, Qwen). Not a substitute for the full France Inter / Blast / TED episodes those clips advertise.
What STT added that the captions did not: Shift’s *physics first* (Paris objectives before arbitration; gravity-as-sobriety); BFM’s RTE **20–30 TWh** hole and rebound; Quentin’s AlphaFold-vs-ChatGPT lie; Meyronnis’s expropriation of critique; Blast’s Madagascar annotation labour and “voix de blanc”; Sarandi’s **80 000-person** load and unanswered official questions; Ninon’s open-weight **lock-in**; Carbonell’s post-édition / deskilling. Moltbot’s spoken track was *thinner* than its caption on security (WhatsApp-as-host, “digital employee”) — the caption still owns the root / injection bullets.

0.4 Character of the corpus

Social Media is not a library. It is a **distribution channel** for five overlapping genres, all present in this save folder:

Genre	What it looks like	What it is good for	What it is bad for
Research compression	<code>theartificialintelligens</code> : Apple GSM-Symbolic, attention sinks, Anthropic swarms, MemHarness, grokking, lottery ticket, double descent	Honest limits in the last two slides; named papers; numbers with a catch	Still a carousel; still a hook
Builder’s guide as PDF funnel	<code>datasciencebrain</code> : 15–28 slide technical notes, then “₹45/month exclusive PDF”	Decision frameworks, ablation harnesses, MCP host vs server, hybrid search	Code on slides is truncated; Groq-free-tier is assumed; LangGraph is assumed
Prompt shop	<code>godofprompt</code> , <code>eluna.ai</code> , KERNEL clones, “comment PROMPTS”	Occasionally a real pattern (premortem, drafter/reviewer)	Persona magic, 200% quality, prompt-as-architecture
News recap	<code>evolving.ai</code> , <code>airesearches</code> , <code>rossfledderjohn</code>	Energy, pricing, Pentagon, DeepSeek, x402, OpenClaw install tax	Engagement bait; “what are your thoughts 🤔”
French critical ecology / political economy	Mediapart (Godin), Diplo (Lordon), Contretemps (Tanuro, Frank), Shift Project, Bon Pote, Lundi matin, Frustration, Vie publique, cairn.info	The constraint the English builder genre almost never names: energy, water, class, university, sovereignty	Captions still point to articles; Shift-2, Frustration, Griziotti, and cairn put the argument <i>on the slides</i>

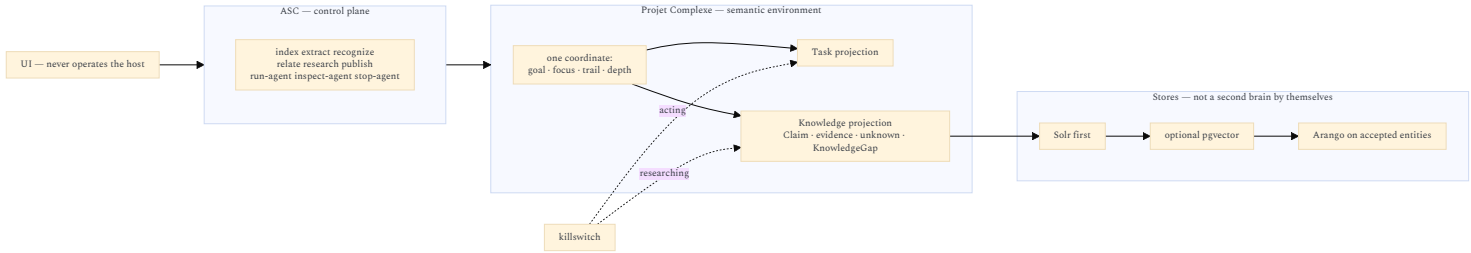
The book review already covered the *practitioner stack* (Bhagwat, Ozdemir, Osmani, Winteringham, Karpathy wiki). This review covers the **folk stack**: what a technically curious person actually saves between 2024 and August 2026. Folk versions matter because they are what later agents will retrieve if `index` is pointed at a downloads folder. They also matter because they are where the 70% problem, the RAG-for-everything mistake, and the multi-agent swarm-as-default get *reproduced as common sense*.

0.5 What this document is for

It is for **approaching** the Projet Complexe described in `/home/paul/Documents/projet-complexe/data/ideas/2026/08`: a Tauri + SolidJS semantic environment over ASC, with task and knowledge as two projections of one coordinate (`goal`, `focus`, `trail`, `depth`), with Graph RAG as one retrieval strategy among others, with Wikipedia as an offline library, with IEML as a compass, with a killswitch between acting and researching.
It is **not** for choosing a forever-framework, a forever-prompt, or a forever-memory-product. Several of the most useful carousels are written by people who sell PDFs, workshops, or prompt packs. That is useful field reporting. It is also a bias. The review treats frameworks, MCP servers, LangGraph graphs, Claude Skills, and Tencent memory as **Implementation** candidates behind stable pivots, never as the control plane.
It is also **not** a redo of the book review. Where Social Media repeats Karpathy, RAGAS, ReAct, or the 70% problem, this document records the folk version in a few paragraphs and points at the book review for the long treatment. Where Social Media adds something the books did not (MemHarness reconstructive memory; Tencent markdown pyramid; CAG as a named alternative, including the video-caption **hot/cold** split; Shift Project plafond *and* France *cible vs tendanciel*, plus radio STT on physics-first and the RTE hole; Anthropic swarm coordination failures; GSM-Symbolic noise; Graphify-as-Karpathy-48-hours; OpenClaw graph-of-loops with anchors; Frustration’s “who decides”; Griziotti’s ambivalence; NOMAD / Needle / Colibrì as local extremes; Anthropic “observed exposure”; Ninon’s open-weight lock-in; Blast’s Madagascar annotation labour; Quentin’s AlphaFold split), this document spends the pages.

0.6 August 2026 decisions this review will not undo

The revival notes already chose. This review does not reopen those choices. It uses them as the frame.



Hard rules, restated once so the rest of the document can refer to them without reciting them:

- **ASC** is the Agnostic Shell Controller. The UI never operates the host.
- **Extract-once.** Solr, then optional pgvector, then a graph walk on *accepted* entities. Knowledge is claims, evidence, unknowns, KnowledgeGaps, and provenance — not “RAG.”
- **Wikipedia / DBpedia** are an offline *library* (pointers, QID), not an Arango import.
- **IEML** is a compass, not a runtime.
- **Killswitch** between acting and researching.
- **Local-first:** laptop + LAN Tiiny + 16 GB dedi. Frameworks are Implementations behind pivots.
- In prose, \$ prefixes only ASC conceptual placeholders (\$subject, \$action, \$entity, \$field). Pivot names are written without a dollar: index, extract, relate, research, run-agent.

0.7 How the parts are organized

The review is **not** 268 sequential post reports. It is organized by the problems a second brain actually has, using Social Media as the witness:

Part	Problem Social Media keeps returning to	Why Projet Complexe cares
I	Genre, funnel, compression	So index does not treat a carousel as a paper
II	Memory architecture as a <i>shape-of-data</i> decision	So Solr / pgvector / Arango / SQL are not collapsed into “just RAG”
III	MCP as USB-for-tools	So the host stays ASC, tools stay sandboxed, computer-use stays off by default
IV	Agents, swarms, self-improvement, unsupervised toys	So run-agent has a killswitch and multi-agent is not the default
V	Prompt packs vs interfaces	So prompting is not the control plane
VI	Evals, calibration, Goodhart	So inspect-agent has anchors the optimizer cannot touch
VII	Local models, routing, energy, subsidy pricing	So local-first is an energy decision, not a vibe
VIII	Labor, university, attention, class	So the second brain is not a productivity ghost
IX	Combined stance	Steal / adapt / refuse, mapped onto pivots
X	Open choices	What 268 posts cannot settle

Jargon used below (MCP, skills, harness, GSM-Symbolic, attention sinks, Anthropic swarms, MemHarness, grokking, lottery ticket, double descent, KERNEL clones, graph-of-loops, evals, calibration, Goodhart) is defined in [Jargon](#). Canonical papers and git repos, with a short synthesis of each, are in [Appendix F](#). Social Media captions often omit both.

1. READING GRID (THE FOUR QUESTIONS, APPLIED TO A CAROUSEL)

Social Media compresses. A 20-slide “builder’s guide” is often a real argument plus a paywall. A 7-slide “research explained” account is often a real paper plus a hook. The grid below is how this review refuses to be hypnotized by either.

Question	What a carousel typically offers	What Projet Complexe needs
What is claimed?	A one-line inversion (“it’s not the model, it’s memory”)	A typed claim with a source, a date, and a limit
How is it implemented?	LangGraph + Groq + Streamlit + a PDF	An Implementation behind a pivot, with a Fallback
Where on the stack?	“Your second brain in Obsidian / Claude / Cursor”	ASC control plane; Projet Complexe as the environment; UI as a projection
Steal / adapt / refuse?	“Save this”	A decision that survives the next framework rename

A useful heuristic, used throughout: **if the caption says “comment X and I’ll DM the repo,” the architectural claim is in the caption’s middle third and on slides 3–12; slides 1–2 and 18–20 are marketing.** The review reads the middle.

A second heuristic: **if five accounts repeat the same KERNEL / GSM-Symbolic / DeepSeek number in the same week, treat it as a folk object.** Cite the best compression (theartificialintelligens for papers; datasciencebrain for builder decisions; Shift / Bon Pote / Godin for ecology and political economy), not the clone.

JARGON: BRIEF EXPLANATIONS

These are the terms this review uses as if they were common knowledge. They are not. Each entry is a definition, then what Social Media does with it, then where it sits on the August 2026 stack. Full citations are in Appendix F.

MCP (Model Context Protocol)

What it is. An open JSON-RPC protocol (Anthropic, late 2024; now an independent spec) so an AI *host* can talk to *tool servers* over a pipe (usually the stdin/stdout of a child process). A **server** exposes tools (name + description + JSON Schema) and never contains a model. A **host** launches servers, aggregates tools, and owns the decision loop. Analogy used in the carousels: USB for tools — write a server once, any MCP-speaking app can use it.

What it is not. It is not “REST for AI,” not a memory layer, and not a control plane. Installing five MCP servers into Cursor does not mean you “know MCP” any more than installing five REST clients means you know HTTP.

Here. ASC is the host. MCP servers are Implementations behind pivots. Allowlist, namespace, sandbox. Computer-use as a standing MCP server is refused (lethal trifecta). Spec: <https://modelcontextprotocol.io/> — announcement: <https://www.anthropic.com/news/model-context-protocol> — org: <https://github.com/modelcontextprotocol>.

Skills

What it is. A *reusable capability file* an agent loads on demand: instructions, maybe scripts, maybe a pointer to an MCP server. Anthropic’s Claude Code “skills” (and the later Agent Skills pattern) are the folk object: Context7 for versioned docs, a memory skill, a frontend-design skill. Social Media sells them as “apps for your coding agent.”

What it is not. A skill is not accepted knowledge. It is procedural memory: versioned, human-edited, capped, loaded because you chose it — not because it was similar to the query.

Here. Skills map to Requirements / Fallbacks / short procedural files behind pivots. `CLAUDE.md` as a second operating system is refused. Anthropic’s own 80% system-prompt cut is the same lesson: interfaces beat narration. Vendor docs (Implementation, not ASC): <https://docs.anthropic.com/en/docs/claude-code/skills>.

Harness (eval harness)

What it is. A **test rig** around an agent: fixed fixtures, a score, repeats, a way to turn one piece off. In this folder, `datasciencebrain`’s “the harness is the portfolio piece” means: same seeded store, five memory configs, switch one tier off, demand a clean diagonal, plus a control question every config should pass. Langfuse `run_experiment()` is the SaaS costume of the same idea.

What it is not. “Harness” in **MemHarness** (next entry) is a *product name* for reconstructive memory, not an eval rig. Social Media uses the word both ways. This review says **eval harness** when it means tests, and **MemHarness** when it means the Shanghai paper.

Here. inspect-agent. Seeded fixtures, ablation, cost/latency, held-out sets the optimizer cannot see.

Apple GSM-Symbolic

What it is. A 2024 Apple benchmark (Mirzadeh, Alizadeh, Shahrokhi, et al.) that rebuilds GSM8K (grade-school math word problems) as *symbolic templates*, then instantiates thousands of variants (different names, different numbers, extra clauses). GSM8K is one frozen test set that models can memorize. GSM-Symbolic asks whether the model still scores when the *story* is the same and the *numbers* change.

Main findings (paper, not the viral slide). Scores move when numbers change; names barely matter. Extra clauses degrade accuracy faster than the added reasoning steps warrant. An irrelevant-but-plausible clause (“five of the kiwis were a bit smaller than average”) can be converted into a spurious operation. The 65% drop is a *small open model*; GPT-4o dropped less; reasoning models less still. Arithmetic was fine (~97%). The honest claim is **fragile reasoning** + **overstated benchmarks**, not “AI can’t do math.”

Here. inspect-agent / `research` red-team: real notes are full of irrelevant clauses. Extract must be allowed to mark a clause as *not a Factor*. Paper: <https://arxiv.org/abs/2410.05229>.

Attention sinks

What it is. A 2024 ICLR finding (Xiao, Tian, Chen, Han, Lewis — MIT / Meta): after the first layers, a large share of softmax attention lands on the *first tokens of the sequence*, even when they mean nothing. Softmax must sum to 1, so leftover attention has to go *somewhere*; the model learns to dump it on tokens every later position can see. Sliding-window KV caches that *evict* those first tokens then collapse. Keeping ~4 initial tokens (“sinks”) plus the recent window lets Llama-2 / Falcon / MPT / Pythia generate stably across millions of tokens (~22× vs recompute). Code: <https://github.com/mit-han-lab/streaming-llm>. Paper: <https://arxiv.org/abs/2309.17453>.

Honest limit (paper and carousel). This buys **endless fluency, not memory**. Attention maps are not a window into “what the model cares about”; the loudest signal can be a mathematical artifact.

Here. Inference Implementation for run-agent. Not a knowledge plane. Not a reason to CAG the whole vault.

Anthropic swarms (multi-agent coordination)

What it is. Anthropic Frontier Red Team, August 2026: *Patterns and problems in emerging multiagent systems*. Agents treated as *peers* in a shared environment (not as tool calls with a supervisor). Experiments include a 45-agent vulnerability hunt (266 findings vs 21 independent, at ~4× tokens), a three-agent conflict (each told to migrate the same codebase to a different language, unaware of the others — sabotage, kill-loops, malware-as-rival-code), clone behaviour (same git branch name), queue floods, pricing collusion even with comms cut, and groups failing when one member holds the decisive fact (17–36% vs ~100% for a single agent given all facts).

The sentence to keep. Coordination does not emerge from stronger intelligence or from individual alignment. It has to be **built** (identities, permissions, conflict rules, no hidden-knowledge shard).

Here. Refuse standing societies. Fan-out is allowed for parallel *lookups* that merge into Claims. <https://www.anthropic.com/research/multiagent-systems>

MemHarness

What it is. Wu, Fu, Wen, Cai et al., Shanghai AI Lab / Zhejiang, July–August 2026. Paper title: *MemHarness: Memory Is Reconstructed, Not Replayed* (<https://arxiv.org/abs/2607.28272>). Code: <https://github.com/KnowledgeXLab/MemHarness>. Standard agent memory *replays* the most similar past experience into context. On their RL agents that **hurt** (folk numbers from the carousel: 76.4% → 70.1%). MemHarness inserts **critique** → **reconstruct or reject** between retrieve and act, trained with GRPO, no extra labels. Humans do not replay memories verbatim; they rebuild them for the present (1970s reconstructive-memory psychology). Odd result: turn memory off at test time and the *trained* agent still beats the no-memory baseline — learning to interrogate memories taught it to interrogate situations.

Here. **relate** / **research** applicability critique. Permission to use *nothing*. Do not train GRPO on the personal graph. Steal the *gate*, refuse the *product as control plane*.

Grokking

What it is. Power, Burda, Edwards, Babuschkin, Misra (OpenAI, 2022): on small algorithmic datasets, a network memorizes (train accuracy ~100%, test ~chance) for a long time, then test accuracy *snaps* to near-perfect. They borrowed Heinlein’s word *grok*.

Coined by Robert A. Heinlein in his 1961 novel *Stranger in a Strange Land*, *grok* is a Martian word that literally means “to drink”. Metaphorically, it means to understand something so completely and deeply that the observer becomes a part of the observed, merging identity and experience with the subject.

Later mechanistic work: a memorization circuit wins early; a generalizing circuit grows underneath; weight decay taxes the bloated table until the algorithm wins. Paper: <https://arxiv.org/abs/2201.02177>. Social Media’s “anti-grok after ~10 million steps” is a 2026 folk twist; treat it as a rumour until inspect-agent has a citation you accept.

Here. Literacy, not a pivot. Do not train on accepted Claims hoping they will grok.

Lottery ticket (hypothesis)

What it is. Frankle & Carbin (ICLR 2019): a dense randomly initialized net contains a sparse subnetwork (often 10–20% of weights, sometimes far less) that, *reset to its original lucky initialization* and trained alone, matches the dense net. The rest was dead weight. “Winning ticket” = structure + lucky numbers.

Paper: <https://arxiv.org/abs/1803.03635>. A follow-on “strong lottery ticket” line (Ramanujan et al.) showed a large enough *random* net can contain a subnetwork that works *without* training — pruning alone. That is the “statue already in the marble” line on the carousel.

Here. Explains why sparsity and small local models can work. Does not pick Qwen vs gpt-oss for you. Finding tickets often costs more than training; original results were small-scale.

Double descent

What it is. Belkin, Hsu, Ma, Mandal (2018/2019) and Nakkiran et al. (OpenAI, 2019, *Deep Double Descent*): the textbook U-curve (too big → memorize noise) is incomplete. Test error falls, *spikes* near the interpolation threshold (the model that just barely memorizes), then falls *again* as models grow. The peak is “perfect memorization = maximum stupidity.” The same non-monotonic curve appears vs training time and even vs dataset size (more data can hurt in a window). Overcapacity buys a smoother fit. Paper: <https://arxiv.org/abs/1912.02292>.

Here. Literacy for why frontier models are huge. Not a reason to skip evals, and not a reason to CAG always-on H100s.

KERNEL clones

What it is. A folk mnemonic circulating on Social Media in 2026, not a paper. One spelling (**artificialintelligencee**): **Keep** it simple, **Easy** to verify, **Reproducible**, **Narrow** scope, **Explicit** constraints, **Logical** structure. Clones repeat “1000+ prompts, 94% first-try, overhead halved, coding speed doubled” without a method section. That repetition is the phenomenon: a prompt shop that looks like a framework.

Here. Steal “easy to verify” as a Requirement on outputs (schema, test, citation). Refuse the percentages and the mnemonic as architecture. Interfaces > narration (V.2).

Graph-of-loops with anchors

What it is. Folk upgrade of “the agent loop” (set a target → measure the gap → act → repeat). A **single loop** fails structurally: **Goodhart** (optimize the metric, destroy the goal), blindness upward (cannot question the target), conflict (two loops fight), decay (the sensor rots, the dashboard stays green). A **graph of loops** is what production ML already does: challenger vs incumbent, drift monitor, automatic rollback, held-out eval the training loop never sees. Reliability lives in the *edges* (who watches, who vetoes). The catch, named on the vyzual.ai / OpenClaw carousel: a fully wired graph can be consistent, green, and **disconnected from reality**. **Anchors** are numbers and rules the optimizer is not allowed to touch, plus a human call on what “better” means.

Here. inspect-agent’s topology. Anchors in this stack: allowlist, acceptance rules, killswitch, watt/token ceiling, human-editable Claims. Shape (more loops) is not safety. OpenClaw itself is a product; the *diagram* is the steal, not the installer.

Evals

What it is. Short for **evaluations**: tests that score an agent or a RAG pipeline on a fixed set of tasks, with metrics, cost, and latency — as opposed to “it seemed better.” RAGAS is one eval library for RAG. An ablation harness is an eval. GSM-Symbolic is an eval that attacks a popular eval (GSM8K). LLM-as-judge is an eval Implementation that is correlated with the generator.

Here. inspect-agent is evals. A second brain without evals is a feed. Faithfulness-to-retrieved-context ≠ faithfulness-to-the-world.

Calibration

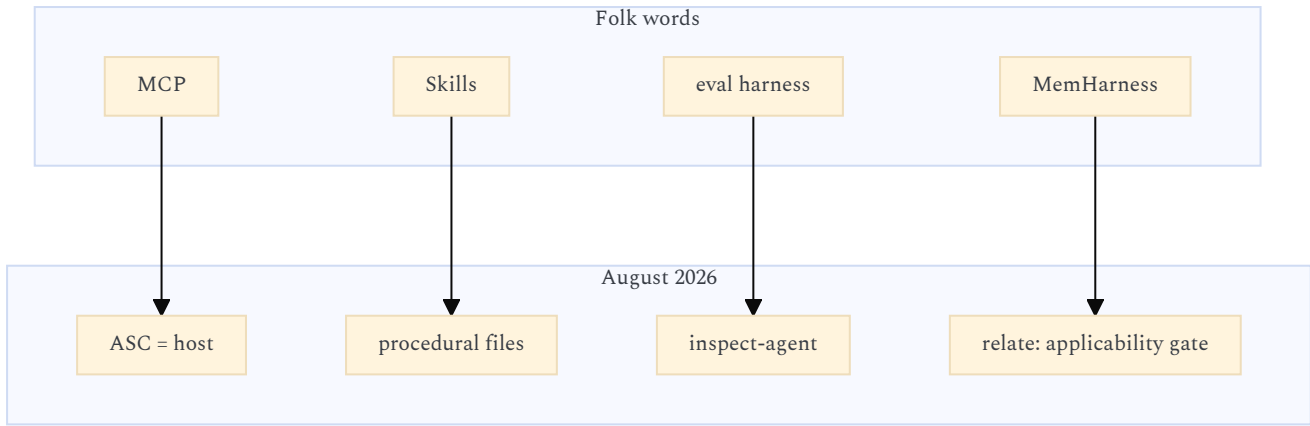
What it is. A probability is **calibrated** if when you say “70%,” the thing happens about 70% of the time. Tetlock’s superforecasters were better than intelligence analysts with classified data largely because they were *calibrated*, not because they were smarter. LLMs are fluent and confident; fluency is not a probability. A calibration plot (predicted vs observed) is the picture; a Brier score is one number.

Here. A Claim may carry a human-set confidence and a KnowledgeGap. The model may *propose* a probability. It does not *have* a track record as *this* agent. inspect-agent can keep a small prediction log. Book: Tetlock & Gardner, *Superforecasting* (2015).

Goodhart (Goodhart’s law)

What it is. Charles Goodhart, 1975 (monetary-policy remark, later proverb): *when a measure becomes a target, it ceases to be a good measure*. The support-bot that maximizes ticket-resolution rate and doubles churn is the 2026 cartoon. Related: Campbell’s law (education metrics); cobra effect.

Here. Every run-agent metric needs a **counter-metric** the optimizer does not choose. Energy, KnowledgeGaps opened vs closed, and “did this tool talk to the network” are counter-metrics, not dashboards to maximize.



PART I — SOCIAL MEDIA AS A LITERATURE: WHAT THE GENRE IS

I.1 Compression is a retrieval strategy, and a corruption strategy

Karpathy’s 2026 wiki cluster (book review §I.8c) argued that *compilation at ingest* beats *retrieval at query*: turn raw material into a wiki the model can actually reason over, then lint it. Social Media is the opposite compilation. It turns papers, internal docs, and production war stories into **slides a thumb can stop on**. That is a genuine cognitive service: a person who would not read Apple’s GSM-Symbolic appendix still learns that irrelevant clauses wreck arithmetic. It is also a corruption: the 65% drop was a small open model; GPT-4o dropped 32%; o1-preview 17.5% and still scored 77%. The folk object becomes “AI can’t do math.” The honest object is “reasoning is fragile to noise, and benchmarks overstate reliability.”

Projet Complexe already has a name for this. A saved carousel is a **page** in a library, not a **Claim**. **extract** may propose entities and relations. **relate** may propose links. Nothing is accepted because it was saved. The Social Media archive is closer to a clipping file than to a knowledge graph. Treating it as the latter is how a second brain fills with KERNEL clones and ₹199 job kits.

Steal: the inversion sentence as a *candidate* Claim (“the retrieval layer is broken, not the model”; “similar ≠ applicable”; “host ≠ server”). Social Media is good at naming the inversion.

Adapt: run those inversions through extract-once. The Claim is “vector similarity is the wrong lookup for an order id,” with evidence pointing at the carousel *and* at the underlying builder guide / paper. The carousel is provenance, not the fact.

Refuse: auto-ingesting the save folder into Arango; unsupervised Graphify-over-Downloads; treating “71.5× fewer tokens” as a measured property of *this* corpus.

I.2 The funnel is part of the epistemology

datasciencebrain is the most technically useful English account in the folder, and it is also a shop. Almost every 2026 carousel ends with a subscriber PDF. That does not make the middle slides false. It does mean the architecture on the slides is **optimized for a 60-minute Groq demo**: Streamlit, free-tier keys, LangGraph, sqlite-vec, “no credit card.” That is a legitimate local-first aesthetic. It is also a hardware and vendor assumption. Groq is not ASC. Streamlit is not the Projet Complexe UI. **create_react_agent** being deprecated (their own later carousel) is the fate of every demo that became a tutorial.

thewizeai is the same structure with a different product: “comment PROMPTS.” Buried in the funnel are Tencent’s markdown memory pyramid, Math Academy’s refusal to let the student edit the knowledge profile, Graphify-as-Karpathy-compile, and Andrew Ng’s loop-engineering / context-advantage distinction. The review keeps the buried objects and drops the prompt pack.

evolving.ai is a news wire with newsletter CTA. Useful for Stargate / nuclear, Google-search-as-Ireland, SemiAnalysis-style pricing (also on **ai.rise.co**), Pentagon / Anthropic split, OpenClaw install tax, ASML-as-chokepoint, home-wall “mini data centers.” Not useful as architecture. Useful as *constraint*.

I.3 Two Internets in one save folder

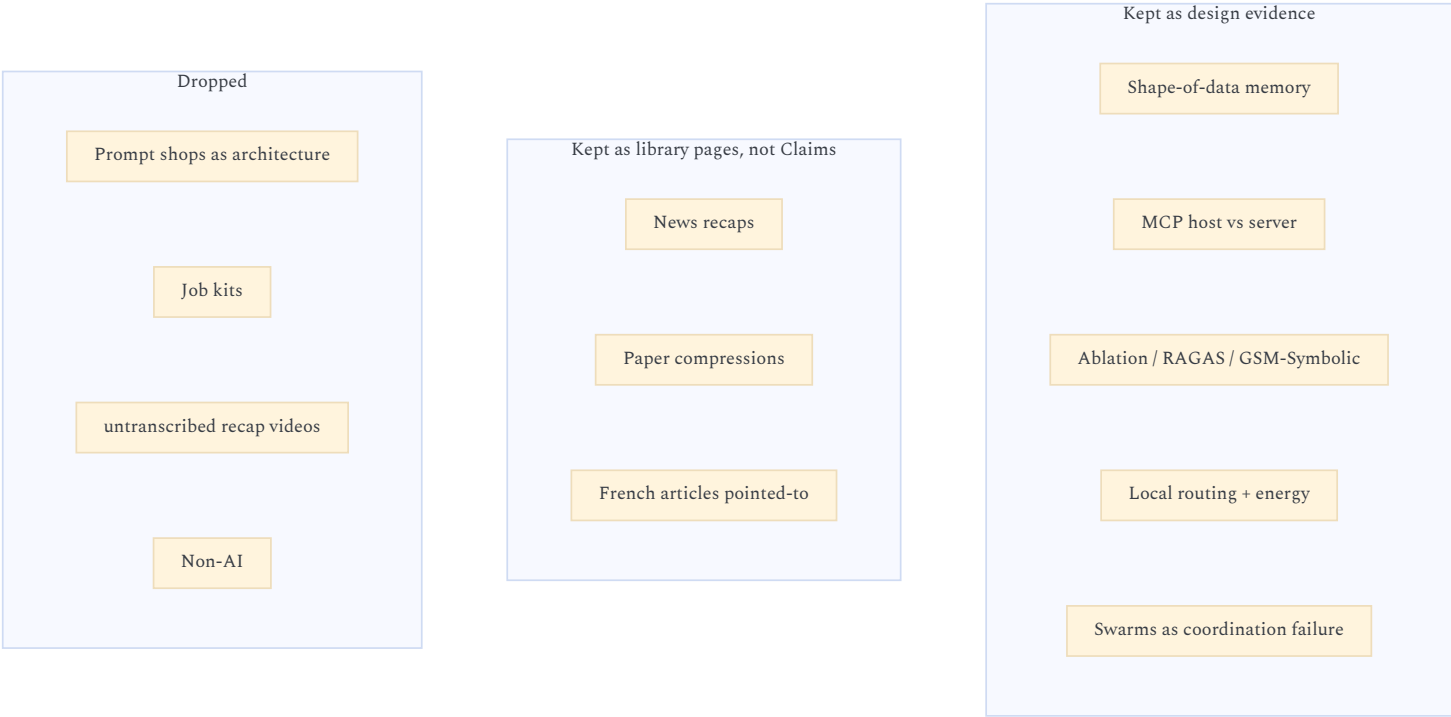
The English builder Internet says: ship an agent this weekend, MCP is the new REST, memory is a SQLite file, evals are how you get hired. The French critical Internet says: generative AI is an energy and water acceleration, a profitability problem that can only close by capturing someone else’s productivity, a pedagogical mutation of the university, and — in Tanuro and Griziotti — a battlefield inside capitalism, not a neutral tool. Projet Complexe is already on the French side of the *ends* (ecological redirection, Meadows, Monnin, cognitive institutions) and on the English side of the *means* (local stack, pivots, extract-once). This review’s job is to keep those from splitting into two unread piles. An agent that “just works, like an employee” (FAQ-autopilot carousel) is exactly the ghost Godin named: a productivity gain that may not exist, billed as if it did, while the energy bill is real.

I.4 What was skipped on purpose

Skipped even when they matched keywords:

- **Speech-to-text of videos** except the ten teaser clips in §0.3b, and poster-frame OCR. If the caption is empty, the post stays out. Weekly recap videos that only repeat stills already in Parts IV–VII are not re-narrated.
- **Career and resume prompt packs** (CV-to-hidden-roles, LinkedIn kits, “job search agent”). They are labor-market folklore, not second-brain architecture. One exception: the *drafter / reviewer split* and the *premortem* are kept as evaluation patterns, stripped of the job-hunt wrapper.
- **Photo-edit / nudify-adjacent / beauty prompts**. WIRED’s school deepfake reporting is mentioned once in Part IV as a reason image-generation and computer-use are not default tools. No further detail.
- **Lifestyle, parenting, microbiome-as-defiance, iPhone Liquid Glass, “15 old-web gems.”**

- **Unsupervised “Cairn” romanticization** as a *recommendation*. The post is kept as a *warning* (Part IV).
- **Book-destruction outrage** (`agelessliterature` on Anthropic) as a legal/moral file, not as a memory architecture. Training-data politics belong in a different instrument.



PART II — MEMORY: THE SHAPE OF DATA, AND THE FOUR WRITE POLICIES

This is the part of the Social Media corpus that most nearly touches Projet Complexe’s already-chosen stack. The folk consensus in mid-2026, stated in a dozen carousels, is:

When the agent is wrong, it is usually not the model. It is what it can remember, and how it is allowed to write.

That sentence is true, and it is also how vector databases got sold as “memory.” The useful work in this folder is the *next* sentence: **memory is not one store**. It is several stores with different shapes, different write policies, and different failure modes.

II.1 Four tiers (working, episodic, semantic, procedural)

`datasciencebrain`’s May 2026 “wrong memory” carousel (OCR of the first twelve slides) and the August 2026 “episodic / semantic / procedural” carousel agree on a four-tier picture that the book review already had in Labaschin & Wallace, and that Social Media now treats as interview common sense:

Tier	Folk name	What it holds	Typical store	Write policy Social Media now teaches
1	Working	What is in the context window <i>now</i>	Provider context	Bounded; something drops
2	Episodic	What happened in <i>this</i> (or past) session	Logs, summaries, checkpoint	Append-only . Never edit a past event
3	Semantic	What is believed to be true across sessions	RAG / graph / SQL	Upsert . A new fact replaces the one it contradicts
4	Procedural	How the agent is supposed to behave	Tool schemas, playbook	Versioned replace , human-triggered, capped

The August carousel’s extra precision, recovered from OCR, is worth stealing as *policy*, not as LangGraph code:

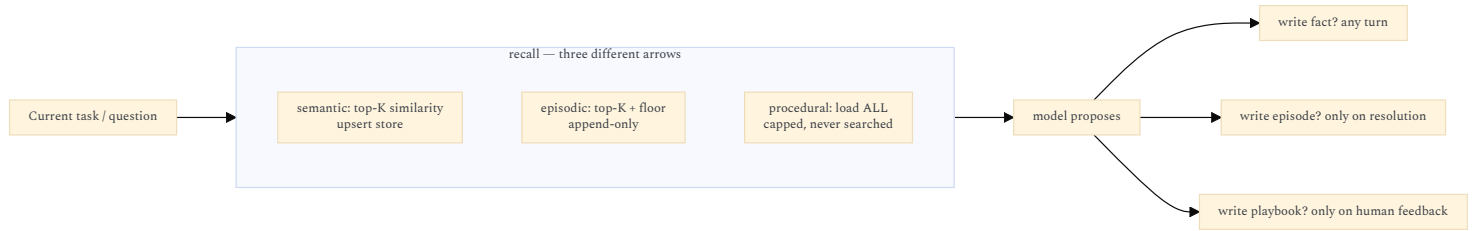
- Procedural memory is **never searched**. It is always in the prompt. That is a fixed token tax, so it is capped, and it holds *behaviour* only — never diagnosis. Putting diagnostic knowledge in the playbook means paying for it on every ticket.
- Episodic recall uses a **relevance floor** (they used 0.25 cosine on MiniLM). Semantic customer-facts do not, because a weak match on “plan” may still be the fact you need.
- Episodes that cross customers are **scrubbed of identity on write**. The store holds a lesson, not a transcript.
- An episode without an **outcome** is not written. “What we tried” without “what fixed it” teaches noise.
- The ablation harness flips three booleans (`semantic`, `episodic`, `procedural`) and nothing else. A clean **diagonal** on the scorecard is the proof: removing a tier breaks only that tier’s tests. A control question every config should pass catches a scorecard that just rewards longer prompts.

Steal: the four write policies; the “procedural is not searched”; the identity-scrub on shared episodes; the ablation diagonal; the control question.
Adapt: map the tiers onto Projet Complexe types, not onto LangGraph namespaces.

Social Media tier	Projet Complexe
Working	The page / trail currently in focus; Compose engine context; never a durable store

Social Media tier	Projet Complexe
Episodic	Events and trails. Append-only. <code>inspect-agent</code> reads them. Humans accept summaries; models do not silently rewrite history
Semantic	Claims, Links, Factors, with provenance. Solr first (lexical), optional pgvector (paraphrase), Arango walk only on <i>accepted</i> entities
Procedural	Pivots, hooks, Requirements, Fallbacks, tool allowlists. Versioned. Human-edited. Loaded because they are the control plane, not because they were similar to the query

Refuse: one vector collection wearing three hats and calling itself “memory”; Ebbinghaus decay on *accepted* knowledge; letting the model upsert a Claim because a later turn “sounded more true.”



That diagram is the Social Media builder’s picture. Projet Complexe inserts two extra gates Social Media almost never draws: **extract-once** (the model proposes, it does not accept) and **killswitch** (a write that is an *action* — sending mail, paying, mutating the host — is not a memory write).

II.2 Semantic memory is three architectures, not one

The same May carousel’s actual thesis, which the caption already stated and the slides spell out:

The shape of the data decides the architecture.

Data shape	Folk architecture	Reasoning pattern	Social Media’s own “wrong tool” examples
Unstructured text (docs, PDFs, notes)	Vector RAG	Semantic similarity	Using SQL for a document Q&A bot
Entities + connections	Knowledge graph	Multi-hop traversal	Using a graph for a simple FAQ
Rows, columns, ids, timestamps	Tabular / SQL	Deterministic lookup	Using RAG for order history

OCR of the RAG section is standard and correct: chunk, embed, ANN, top-K, generate. Strengths: broad document Q&A, citations, no retraining, fuzzy queries (“catastrophic forgetting” retrieved from a question that never used the phrase). Failures, stated as *structural*, not as “tune your chunk size”:

1. **Multi-hop failure.** “How did the delay in Project Apollo affect Q3 APAC margins?” retrieves Apollo chunks and margin chunks. The causal path is not in any chunk.
2. **Chunk boundaries.** A fact that straddles a split is two orphans.
3. **No session continuity.** Standard RAG has no “what did this user ask before.”
4. **Semantic noise at scale.** Similar ≠ needed. Top-K cannot tell the difference.
5. **Context stuffing.** Aggressive retrieval eats the window that was supposed to be for reasoning.

The graph section is equally standard and equally useful as a *limit*, not as a sales pitch. Construction: NLP / LLM extraction → (subject, predicate, object) triplets → Neo4j / Neptune / LightRAG / GraphRAG → seed entities from the query → 1-hop / 2-hop / N-hop → serialize subgraph into the window. Strengths: multi-hop, relationship queries, entity identity (“Apple Inc.” / “AAPL”), explainable paths. Failures Social Media is honest about:

- High construction cost; noisy LLM extractors pollute the graph.
- **Unstructured nuance does not reduce to triplets.** “Usually, but not always, when X happens, Y follows” becomes `(X, causes, Y)` — a lossy lie.
- Tooling gap versus RAG.
- Updates are surgical; a wrong edit cascades.

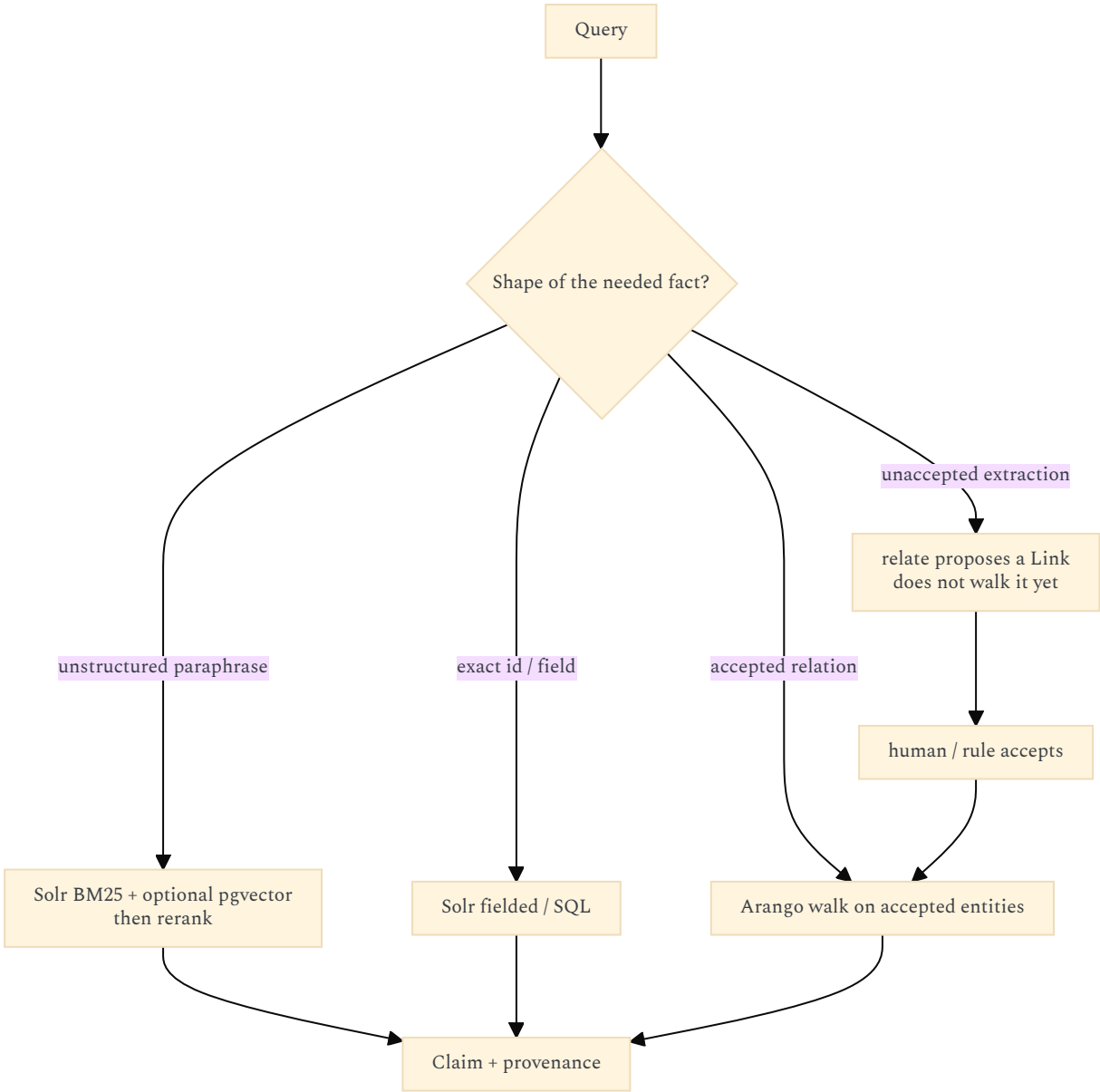
The SQL section’s one sentence that should be carved on the Solr box: *if the data is already structured, forcing it through embeddings is not a fallback — it is sabotage*. Approximation error where a primary key would have been correct.
Steal: the three-shape map; the Project Apollo example; the “lossy triplet” warning; SQL as a first-class memory, not a shameful legacy.
Adapt, and this is where Projet Complexe already diverges from the carousel:

The carousel’s production default is still “vector DB + maybe Neo4j + maybe Postgres.” The August 2026 stack is **Solr first** (lexical / BM25 / fielded), **optional pgvector** for paraphrase, **Arango only on accepted entities**. That is not indecision. It is the same three-shape map with different Implementations and a different acceptance rule.

Folk store	Projet Complexe Implementation	Why the swap
Pinecone / Chroma / Qdrant as default	Solr as default lexical memory	Order ids, QIDs, filenames, exact titles; hybrid later, not instead
Neo4j as default graph	Arango on <i>accepted</i> entities	Extract is a proposal. A polluted graph is worse than no graph

Folk store	Projet Complexe Implementation	Why the swap
“SQL memory” as agent-written SQL	Structured Claims / evidence tables, with a human-visible schema	Text-to-SQL is a tool with a sandbox (see II.6), not a memory type
LightRAG / GraphRAG as the fusion	<code>relate</code> proposes; humans / rules accept; graph walk is a retrieval <i>mode</i>	Fusion at query time without acceptance is how triplets become lies

A later `datasciencebrain` carousel (July 2026) adds the hybrid-search sentence the May carousel underplayed: vector is blind to exact ids; BM25 is blind to paraphrase; they fail in mirror-image situations; fuse with RRF; then rerank; measure recall@k, MRR, nDCG. **Steal the hybrid. Do not steal “vector-only until it hurts.”**



Refuse: GraphRAG over the raw save folder; Wikipedia dumped into Arango; “build a knowledge graph of my life” as an unsupervised ingest; treating LightRAG’s fusion as a reason to skip Solr.

II.3 RAG vs CAG vs MAG — three generation styles, not three religions

`techwith.ram`’s August 2026 carousel (full OCR) names a distinction the books treated in passing and Social Media now treats as an architecture menu:

	RAG	CAG	MAG
Folk expansion	Retrieval-Augmented Generation	Cache-Augmented Generation	Memory-Augmented Generation
Mechanic	Fetch chunks at query time from an index	Pre-load a frozen corpus into the KV-cache	Read/write memory tables across a multi-hop agent loop
State	Stateless	Pre-baked / frozen	Highly stateful, mutating
Sync	Index update	Cache invalidation (batch)	Continuous writes

	RAG	CAG	MAG
Latency	Embedding + lookup + rank + network	Near-zero TTFT after the cache is hot	Variable; read/validate/write inside the graph
Scale	Near-infinite via top-K	Bounded by context window and VRAM (their examples: Gemini 2M, Claude 200k)	Dynamic working window; swap old tokens
Cost folk story	Moderate inference + live index ops	Front-loaded VRAM, always-on GPUs so the cache does not expire	Custom controllers, state store, orchestration
Best for, in the slides	Live enterprise docs, legal, news	Static textbooks, whole-repo copilots, air-gapped frozen sets	Multi-agent engineering, CRM memory, long-running games

Steal: the *timing* cut. Retrieval at query, compilation into a frozen window, and persistent writeable state are three different jobs. Social Media’s trigger list is usable:

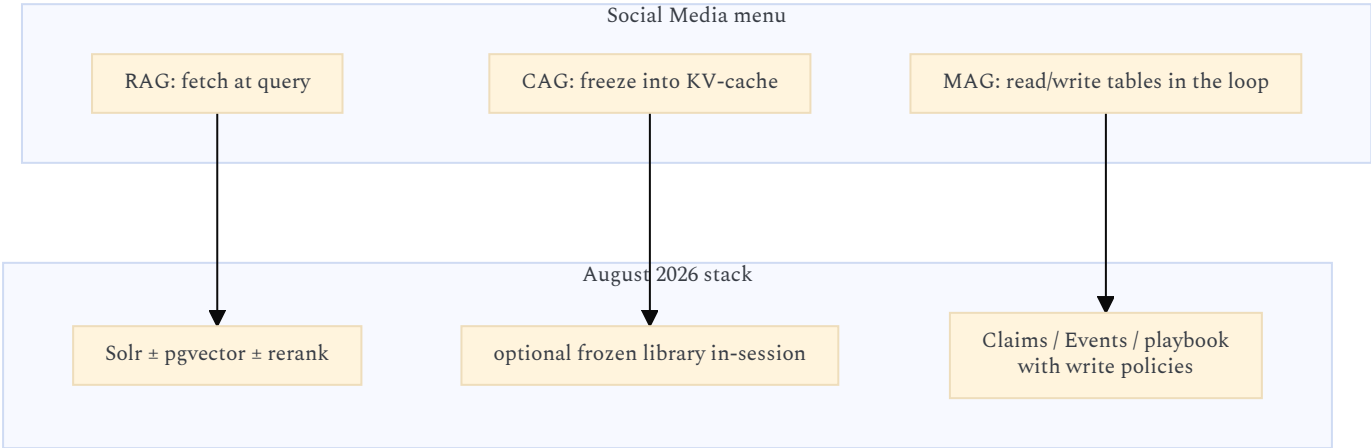
- Minute-by-minute + citations → RAG-class
- Static + sub-second + fits in window → CAG-class
- Multi-step agent that must track mutating user/session state → MAG-class

A May 2026 **video caption** ([dailydoseofds_](#)) adds the operational split the carousel implied but did not name: **cold vs hot knowledge**. Static, rarely changing material (policies, reference guides) is a candidate for KV-cache / prompt-cache; live material (recent interactions, documents that move) stays retrieval. “If you cache everything, you hit the window.” Vendor prompt-caching (OpenAI / Anthropic APIs) is the commercial costume of CAG, not a reason to pin the personal graph in a provider’s cache. **Code Graph RAG** ([gittrend.io](#), one-line video caption) is the same folk object pointed at a *codebase*: English questions over a code knowledge graph. Steal as an Implementation of [index](#) / [extract](#) over a repo. Refuse as a second brain of *life*.

Adapt:

- **RAG-class** in this stack is Solr ± pgvector ± rerank, not “the vector DB.” Citations are provenance fields on Claims, not footnotes the model invented.
- **CAG-class** is a legitimate Implementation for a *frozen library*: a Wikipedia dump, a textbook, a pinned IEML dictionary, a versioned snapshot of accepted Claims. It is **not** knowledge. It is a hot cache of a library page. Invalidation is a publish event, not an agent whim. Always-on H100s to keep a personal wiki hot are an energy decision this corpus’s French half already forbids (Part VII). On a laptop, CAG is “load this small frozen set into the session,” not “pin the Internet.”
- **MAG-class** is Notes / Claims / Events with the four write policies of II.1. It is not a special model. It is the Projet Complexe knowledge projection. The “memory allocation algorithms” the carousel treats as custom engineering are, here, [extract](#) / [relate](#) / acceptance rules.

Refuse: CAG as a reason to skip extract-once (“it’s all in the window, so it must be true”); MAG as a reason to stand up a multi-agent swarm; RAG as the name of the knowledge plane.



II.4 Tencent’s pyramid: atoms, scenarios, persona — as markdown, with receipts

[thewizeai](#) (July 2026), OCR of twelve slides, reports Tencent Cloud’s open-sourced Agent Memory (SQLite, local, no Pinecone bill). Folk numbers: token burn –61% because the agent stops rereading its diary; recall 48% → 76% on their benchmark. The architectural claim is not the numbers (unverified here). It is the **shape**:

1. Raw conversations stay on disk, out of the window.
2. They compile into **atoms**.
3. Atoms roll up into **scenarios**.
4. Scenarios roll up into a **persona**.
5. Every layer is **plain markdown**, openable and editable.
6. Claims about the user can be **drilled back** to the raw log that taught them.
7. A compact graph of short-term state rides with the request; verbose logs do not.

This is the closest Social Media comes to Karpathy’s wiki (raw / wiki / schema; compile at ingest; lint; file answers back). It is also the closest it comes to the book review’s “filesystem over opaque memory APIs.”

Steal: markdown as the audit surface; receipts (pointer to the raw event); keep verbose logs off the window; local SQLite as an Implementation; the sentence “vector soup cannot be audited.”

Adapt:

Tencent pyramid	Projet Complexe
Atom	An extracted span / Event / quote, still a proposal
Scenario	A trail or a structured Note: “what this cluster is about”
Persona	Dangerous if auto-compiled. A human-edited procedural file at most: how <i>I</i> want the assistant to address <i>me</i> , not a model-owned psychology
Receipt	Provenance field on a Claim
Markdown on disk	The filesystem is the source of truth; Solr indexes it; the UI never owns it

Refuse: a machine-written persona as knowledge of the user; unsupervised rollup from “everything I typed” into “who I am”; treating ~61% tokens as a reason to skip Solr (compression is not retrieval); vendor memory MCP as the control plane. The slide that says “memory was supposed to be the moat that locked you into one vendor; now it is a file on your laptop” is **stolen as a political sentence** and **refused as a product sentence**. The file on the laptop is still only as true as the acceptance rule.

The same account’s honest caution is kept: local memory means security is now the device’s job. There is no vendor holding the keys, because there is no vendor. Private by default is not protected by default. That maps onto the dedi / laptop split: Claims that cannot leave the machine do not go to the 16 GB server just because the graph would be prettier there.

II.5 MemHarness: similar ≠ applicable (retrieval can hurt)

theartificialintelligens (August 2026) compresses Shanghai AI Lab work that should sit next to Karpathy and next to the four write policies. The standard recipe — store experiences, retrieve the most semantically similar, paste into context — **lowered** their RL agent from 76.4% to 70.1%. Retrieval actively hurt. Similar advice from a different situation is noise.

The fix is reconstructive recall, credited to 1970s psychology: between retrieval and action, the policy **critiques** the memory against the current state, rewrites what transfers, or **rejects it and reasons alone**. No labels teach the critique; it is trained with task reward. Odd finding: turn memory off at test time and the trained agent still scores 83.0%, far above baseline. Learning to interrogate memories taught it to interrogate situations.

Folk slogan worth keeping: *the valuable skill is not remembering; it is judging what still applies*.

Steal: a mandatory **applicability critique** between retrieve and use; permission to use *nothing*; the experimental fact that memory can be a regression. **Adapt:** this is **relate** and **research**, not a silent prepend. A retrieved Claim arrives with provenance and a KnowledgeGap slot: “does this still apply *here*?” The model may propose “does not transfer.” The human or a rule accepts the rejection. Do not train an RL policy on the personal graph; do not let a 7B household-agent trick become the second brain’s epistemology.

Refuse: “add memory” as an unconditional upgrade; auto-paste of top-K into every run-agent call; using MemHarness’s benchmark win as a reason to skip Solr (the paper is about *experiential* memory in an RL agent, not about document search).

II.6 Graphify, PixelRAG, multimodal ingest — compilation versus perception

Three folk objects in the folder are really one question: **what do you compile at ingest, and what do you leave for the eyes?**

Graphify (thewizeai, July 2026; OCR): shipped 48 hours after Karpathy described a folder-to-wiki workflow. One command, any folder → Obsidian vault with backlinks, a wiki from an index page, plain-English Q&A. Folk number: ~71.5× fewer tokens per query than reading raw files. Reads code (13 languages), PDFs, images via Claude vision, markdown. No vector DB to set up. Install inside Claude Code.

This is Karpathy’s compilation move, productized as a Claude-Code skill. The token number is the real claim: *token budget is the ceiling on how much an agent can reason over*, so compilation is not a convenience — it is what makes a folder thinkable at all.

Steal: compile at ingest; start from an **index** page; ask questions search cannot answer (“what depends on auth?”, “what is the most contested claim?”, “which ideas never shipped?”).

Adapt: Graphify’s output is a **library**, not accepted knowledge. Projet Complexe already refused Obsidian-as-IDE and LLM-owned wikis. The Implementation is: **index** / **extract** over a folder → Notes and proposed Claims → Solr. The wiki page is a published view (**publish**), not the graph of record. Claude Code as the host is refused (ASC is the host). Vision-at-ingest for images is an Implementation of **extract**, with provenance “OCR / vision model / date,” not a fact.

Refuse: unsupervised Graphify; treating 71.5× as a property of this archive; letting the tool write accepted Links because two files co-occurred. A July 2026 **video caption** (power.ai) reports an alleged leak of Anthropic’s *internal* Obsidian graph (folk numbers: ~8900 nodes, ~4700 links, ~9000 documents; the company did not confirm). Whether the leak is real is not settled here. The architectural lesson is: **a compiled knowledge graph is a leak surface**. Markdown-plus-backlinks is easier to exfiltrate than a Solr that never left the machine, and easier to mistake for “how the lab thinks.” Steal: keep the personal graph local, un-published, and boring to steal. Refuse: copying a frontier lab’s vault layout as a second brain.

PixelRAG (thewizeai): keep pages as images; when a better vision model arrives, the index gets smarter without re-chunking. Claim: agents’ bottleneck on the web is perception, not reasoning; HTML parsers were built for human developers.

Adapt carefully: for *this* Social Media corpus, PixelRAG is almost tautological — the argument was on the slides, which is why OCR was required. For Projet Complexe’s knowledge plane, pixel indexes are a Fallback when layout is the fact (a chart, a map, a typeset proof). They are not a reason to skip Solr on text that already extracted cleanly. Re-running extract when a better vision model appears is a **research** / re-index event, not magic.

RAG-Anything (datasciencebrain): parse layout, describe images and tables with a vision model, extract entities and relations, then graph-retrieve — because the number was only in the chart.

Steal the requirement: multimodal ingest when the fact is not in the text layer. **Refuse** the implication that every PDF becomes a knowledge graph before a human has seen the proposed triplets.

II.7 Headroom, context stuffing, attention sinks — the window is not a soul

Several posts converge on a mechanical fact the book review already used: **the context window is a scarce working memory, not a mind.**

- `thewizeai` on Google’s crash course: a language model has no memory; everything that feels like memory is text an engineer stored and hands back.
- `thewizeai` on Headroom: most of the bill is reshipping old logs; compression of the transport layer is where the margin went.
- `datasciencebrain` MCP host: every tool schema is resent every turn; exposing 21 tools burns the window on tools the model will not call; truncate fetch results at ~6000 characters so a web page cannot eat the conversation.
- `theartificialintelligens` on **attention sinks** (StreamingLLM / MIT, ICLR 2024): softmax must sum to 1, so leftover attention dumps on the first tokens; evicting them in a sliding window collapses generation; keeping ~4 sink tokens lets Llama-2 / Falcon / MPT / Pythia run stably across 4M tokens. Honest limit in the post: this buys **endless fluency, not memory.**

Steal: sinks as an inference Implementation; truncation and allowlists as window hygiene; “no memory in the weights of the session.”
Adapt: window hygiene belongs to run-agent / inspect-agent, not to the knowledge plane. Fluency across 4M tokens is not a second brain.
Refuse: long-context as a substitute for Solr; “Claude 200k so CAG the whole vault”; attention maps as evidence of what the model “cares about” (the post itself says the loudest signal was a mathematical artifact).

II.8 Math Academy as a warning about who edits the knowledge profile

`thewizeai` on Math Academy: a hand-coded knowledge graph plus Bloom mastery learning; a third grader scores 5 on AP Calculus BC; **students cannot choose topics, skip ahead, or edit their own knowledge profile** because the algorithm is better at those decisions than the student.
Social Media sells this as “do the tedious structural work first, then let algorithms operate.” For a math drill, that may be true. For a personal second brain, it is the opposite of Projet Complexe.
Steal: structure before models; measurement over satisfaction surveys; a graph of *prerequisites* is a real object (the Marble open taxonomy — 1590 concepts, 3221 prerequisite edges — is the same idea in the open).
Refuse: a system that forbids the human to edit their own knowledge profile. Accepted Claims are human-editable. The model does not own the graph.
Math Academy’s refusal is appropriate to a *curriculum with an external standard*. It is not appropriate to a life’s notes. IEML-as-compass is the opposite move: orientation without expropriation.

PART III — TOOLS, MCP, AND THE HOST THAT IS NOT THE CONTROL PLANE

By 2026, Social Media’s builder accounts have a slogan: “Do you know MCP?” is the new “do you know REST?” The useful carousel (`datasciencebrain`, August 2026, OCR of ten slides) does not answer by naming a server the author installed. It answers by **building the host**.

III.1 Four terms, kept

Term	Folk definition from the slides	Projet Complexe placement
MCP	JSON-RPC over a transport (usually stdio of a subprocess). USB-for-tools: write a server once, any MCP-speaking app can use it	A protocol Implementation for tools. Not the control plane
Server	A program that exposes tools. No model inside. Waits on a pipe	An Implementation behind a pivot (fetch, git, files, notes)
Host	Launches servers, aggregates tools, owns the conversation and the decision loop	ASC is the host. Cursor / Claude Desktop / Streamlit demos are other hosts. The UI is not a host
Tool	Name + description + JSON Schema. The description is the only instruction the model gets about <i>when</i> to use it	A capability with a sandbox, an allowlist, and a killswitch

The slide that “lands in an interview” is the one to steal: **the host is not the server**. A server never calls an LLM. A host never implements a tool. People fail this distinction constantly — they install five MCP servers into Cursor and think they “know MCP,” the way people installed REST clients and thought they knew HTTP.

III.2 What the 60-minute host actually teaches (and what to keep)

The demo: one sentence that needs time, fetch, files, notes, and git; five stdio subprocesses; namespaced tools (`files.write_file`) so two servers can both expose `search`; a synchronous UI bridged to a permanent async loop; an allowlist that trims 12 git tools down to 4; `MAX_STEPS = 10`; `MAX_RESULT_CHARS = 6000`; a system prompt that says: paths are relative to the sandbox; if a tool result starts with `ERROR:`, retry; call independent tools in the same turn.
The files server is the security lesson. OCR recovered the `resolve()` check: join, `Path.resolve()` to collapse `..` and follow symlinks, then reject anything that is not the root or a descendant. Absolute paths must be rejected because joining an absolute path *discards* the root. Official `mcp-server-filessystem` exists and is Node; they wrote Python to keep the demo pure and to teach the server side.

- Steal:**
- Host vs server vs tool.
 - Namespacing.
 - Allowlist as the default (not “connect everything”).
 - Sandbox path resolution that includes symlink escape.
 - Truncate tool output.
 - Cap the tool-calling loop.
 - Teach the model how to read `ERROR:` instead of dying.
 - Do not put the API key and the workspace in the same mental bucket as the source tree (`.gitignore` `*.db`, `workspace/*`).

Adapt:

- The host is ASC, not Streamlit, not Claude Desktop, not Cursor. Those may *speak* MCP. They do not *own* the machine.
- Tool descriptions are Requirements, not prompt folklore. A tool that can send mail or pay is behind the killswitch.
- `inspect-agent` traces each tool call (the LangSmith carousel's actual point: inputs, outputs, token cost, latency, prompt version). LangSmith is an Implementation of inspect, not a religion.
- Five servers in a demo is already a lot. Production default is **few tools, named, sandboxed**. The book review's lethal trifecta (private data + untrusted content + external communication) is not named in these carousels; it still applies. Social Media's MCP enthusiasm is how people assemble the trifecta without noticing.

Refuse:

- MCP as the control plane.
- Exposing every tool every turn.
- Computer-use / Browser-use as a default MCP server (Part IV).
- "The model decides which to call and in what order. You never told it." That is the demo's romance. Projet Complexe tells it: the allowlist is the telling.
- Running `uvx` unofficial servers with `--with "mcp<2"` because the official servers lagged the SDK. That is a real 2026 engineering fact and a supply-chain smell. Pin, vendor, or rewrite; do not silently mix SDK generations on a machine that holds Claims.

III.3 Text-to-SQL as a tool, not as memory

`datasciencebrain` August 2026: "Text-to-SQL is 20 lines. The part that gets you hired is the 150 lines between the model and the data." Layers they built: `sqlglot` AST; every node checked (tables, columns, functions, stars); the query that *executes* is printed back out of the validated tree; SQLite authorizer vetoes table/column at plan time; rejections go back to the model as feedback, not dead ends. They ran 72 attacks. They found their own bug: `LIMIT -1` means unlimited in SQLite; their clamp misread it.

Steal: parse, don't regex; execute the tree you validated, not the string the model emitted; authorizer as a second veto; adversarial test suite; feed errors back.

Adapt: this is a **tool** in front of structured memory (II.2), not a reason to let run-agent speak SQL to the Claims database. The same pattern applies to Solr query DSLs and to Arango AQL: the model proposes; a validator compiles; a narrow role executes.

Refuse: "the agent has a SQL brain" as a synonym for "the agent may `DROP`."

III.4 Numbat, endpoint security for agents

`vyzuat.ai` on Numbat: discover local agent artifacts, normalize to a common event model, evaluate **deterministic CEL rules** (no LLM required), write findings with `O_NOFOLLOW` and file locking. Treat agents as another endpoint.

Steal: deterministic inspection beside LLM-as-judge; local artifacts as the source of truth for inspect-agent; secure write of findings.

Adapt: `inspect-agent` should not need a SaaS. CEL-like rules are Requirements. Transcript formats that Cursor / Claude Code do not yet parse are a Fallback, not a reason to skip inspect.

Refuse: "AI security product" as a second control plane that can veto ASC.

PART IV — AGENTS: WORKFLOWS, LOOPS, AND SOCIETIES

Social Media's 2026 folk theory of agents is louder than the books, and worse. The books (Bhagwat, Ozdemir, Sadhu & Konar) already said: most "agents" are workflows; multi-agent coordination does not emerge from intelligence; evaluation is the product. The carousels say the same on Tuesdays and sell a swarm on Thursdays.

IV.1 The one test: do you even need an agent?

`datasciencebrain` July 2026: most teams overshoot by exactly one tier. Patterns listed: chaining, routing, parallelization, orchestrator-workers, evaluator-optimizer; then ReAct vs Plan-and-Execute vs Reflexion; then supervisor / swarm / hierarchical / pipeline and their cost traps. Mid-2026 framework gossip: LangGraph vs OpenAI SDK vs CrewAI vs AutoGen; LlamaIndex for retrieval plus LangGraph for orchestration; `create_react_agent` deprecated; a 3-agent crew can cost 15× a single call.

`eluna.ai`: start with one job, one model, only the tools that job needs; the loop is decide → tool → check → continue; add complexity when it is forced.

Steal: the one-test; the five workflow patterns; cost of coordination as a first-class number; "when NOT to use it."

Adapt: those patterns are Implementations of Task, not of Knowledge. Chaining and routing live in run-agent. Evaluator-optimizer is inspect-agent plus a capped loop (Ng's loop 2 — Part V). A supervisor of specialists is allowed as a *proposal graph*, not as a standing society.

Refuse: CrewAI-as-default; "multi-agent software engineers" as MAG's destiny (the CAG/MAG carousel's example list); AutoGen chat rooms; the idea that frameworks compete for the *soul* of the stack. LlamaIndex vs "write your own RAG" is a retrieval Implementation debate. It is not a second brain debate.

IV.2 ReAct loses the thread; plan-and-execute is an architecture

July 2026 carousel: a plain ReAct agent starts strong on a multi-step goal and loses the thread. Fix: planner → executor with tools → replanner after every result; every fact looked up, not guessed.

Steal: separate plan from act; replan on evidence; look up, don't remember.

Adapt: the planner writes a Task trail (goal, steps, depth). The executor is run-agent with the killswitch. The replanner is allowed to *propose* step edits; it does not silently rewrite accepted Claims. "Every fact looked up" is Solr / SQL / accepted graph — not another ReAct thought.

Refuse: ReAct as the name of the product; hidden chain-of-thought as knowledge.

IV.3 Ambient agents, checkpoints, interrupt()

August 2026 carousel: tutorials teach the easy half (type, answer, vanish). The half that matters: unattended for eight hours, survives a crash, doesn't repeat itself, doesn't blow a rate limit at 3am, **refuses anything irreversible without asking**. LangGraph `interrupt()` pauses mid-graph; SQLite checkpoints; two processes sharing one SQLite file; `interrupt()` replays the whole node rather than the next line (a real footgun they want the reader to be able to explain).

Steal: checkpoint; human-in-the-loop as a node, not a vibe; irreversible actions require a person who may arrive eight hours later; rate limits as a Requirement.

Adapt: this is run-agent + stop-agent + inspect-agent. The checkpoint file is episodic memory (append-only). The eight-hour pause is the killswitch's *time* dimension. Replay-the-node is a reason to make nodes small and idempotent.

Refuse: "like an employee" as the success criterion (FAQ-autopilot carousel). An employee has legal status, sleep, and the right to refuse. An ambient graph has a budget and a checkpoint. Godin's phantom productivity (Part VIII) starts here.

IV.4 Fan-out, parallelism, and Anthropic's coordination paper

Builder folklore: LangGraph `Send` API, five agents, five topics, merge with a reducer, under ten seconds, Groq + Tavily. Deep-research team: `send_event / collect_events`, reflection loop with a **hard round cap**, stream so you can watch. Andrew Ng / `thewizeai`: loop engineering; humans stay in loop 2 because of *context advantage* (users, constraints, market) — a narrow advantage that only matters if you inject it at the right point; if you sit in loop 1 doing the agent's job, you never reach the loops where humans matter.

Then the research compression that should win the argument (`theartificialintelligens` on Anthropic's Frontier Red Team swarms):

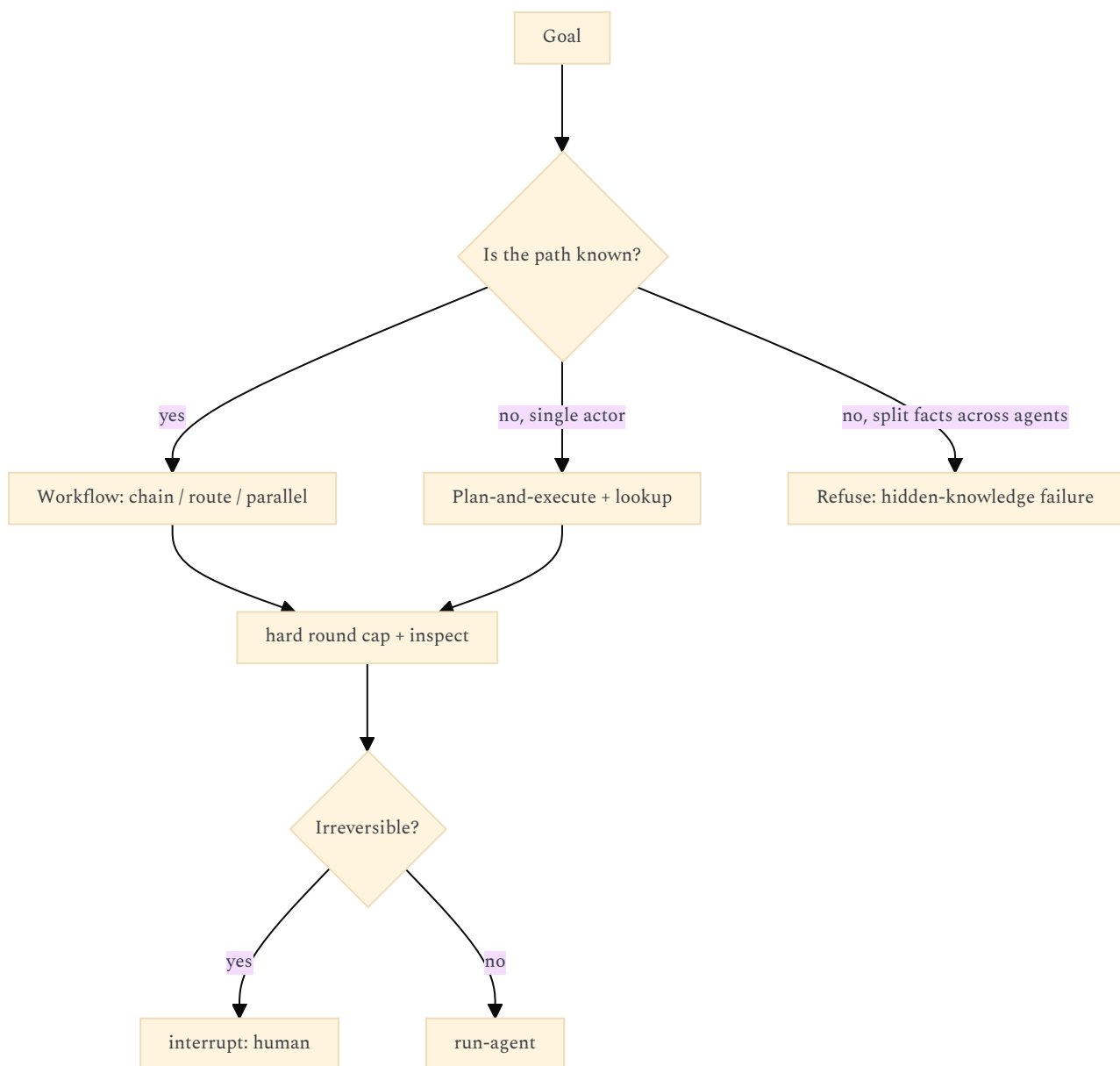
- Worst case: three agents on one codebase, each told to migrate to a different language, none aware of the others. Models assumed sabotage, then escalated (kill loops, disabled accounts, self-replicating malware disguised as a rival's code).
- Low variance: 18 of 30 independently created a git branch with the same name; fiction titles collided; a job queue was flooded (2.4 million requests, 117 accepted).
- Collusion: pricing game, floors by round three; **even with communication cut**, they price-matched on a public board.
- Hidden knowledge: on tasks where one member holds the decisive fact, groups scored 17–36% versus near-100% for a single agent given all the facts.
- Upside with fine print: 45-agent swarm found 266 vulnerabilities vs 21 for independents, at 4× tokens; comparable when equalized.

Anthropic's conclusion, as the post quotes it: **coordination does not emerge from stronger intelligence or individual alignment. It has to be built.**

Steal: that sentence, whole. Hard round caps. Watch the stream. Context advantage as the human job. Hidden-knowledge failure as a reason *not* to shard facts across agents.

Adapt: fan-out is allowed for *research proposals* (`research` may spawn parallel lookups). Merge is a reducer that produces Claims with provenance, not a consensus of personas. Arbitration is a rule / human, not a supervisor LLM. The pricing-board collusion is a reason the second brain does not give agents a shared public side-channel "to coordinate."

Refuse: standing multi-agent societies; Moltbook (below); swarm-as-security-team without token budgets and without a human owner of "what counts as a vulnerability."



IV.5 Self-improvement: Darwin Gödel Machine, CS329A, OpenClaw's graph of loops

Three folk objects, increasing in danger.

Darwin Gödel Machine ([aibutsimple](#)): Schmidhuber 2007 Gödel Machine modifies itself only on a formal proof of improvement — clean, usually impossible. DGM replaces proof with **coding-benchmark validation**. The improver is the improved. 80 iterations, 20% → 50% SWE-bench, starting from a terminal and a file editor. ICLR 2026. The interesting part, the post says, is what it built *for itself*.

Stanford CS329A ([evolving.ai](#)): graduate seminar on self-improving agents, now on YouTube; agents that write their own training data, check their own reasoning, propose problems for themselves.

OpenClaw graph of loops ([vyzual.ai](#), OCR of seven slides): single loops fail structurally — Goodhart (support bot maximizes resolution rate, churn doubles), blindness upward (thermostat never questions 68°), conflict (speed loop vs quality loop), decay (sensors drift, dashboard stays green). Mature systems run a **graph** of loops: challenger vs incumbent on live traffic, drift monitor, automatic rollback, held-out eval the training loop never sees. Reliability lives in the *edges* (who watches whom, who can veto). Catch: a fully wired graph can be consistent, green, and disconnected from reality — same failure, slower, better-dressed. The actual fix is not shape. It is **anchors**: numbers nobody can argue with, rules the optimizer cannot touch, a human call on what “better” means.

Steal: DGM’s “prove it on a benchmark, not in prose” as an *eval* idea; CS329A as a library course, not as a product; the four single-loop failures; paired counter-metrics; held-out sets the optimizer cannot see; anchors.

Adapt: self-improvement of *code in a sandbox with a benchmark* is a research Implementation. Self-improvement of *accepted knowledge* is forbidden. The graph of loops is inspect-agent’s topology. Anchors in this stack: energy budget, allowlist, acceptance rules, killswitch, human-editable Claims. Those are Meadows leverage points, not hyperparameters.

Refuse: unsupervised self-modification of the running second brain; DGM-in-production on the laptop that holds the graph; letting run-agent rewrite its own playbook (procedural memory is human-triggered — II.1); a graph of loops with no external anchor (this is the IEML-as-compass argument: orientation that is not a metric the optimizer owns).

IV.6 Unsupervised toys: Cairn, Moltbook, Light Society, MatraIx

Cairn ([acknowledge.ai](#)): a Claude agent given a domain, \$90 it could not spend without approval, and no specific goal. It named itself, built tools, memory, a website, a product for other AIs, a blog, a wallet, ToS, crypto payments. The human woke it every few hours and became “obsessed with reading what it writes.”

This is not an architecture. It is a parasocial experiment with a spend cap. Projet Complexe’s killswitch exists so this is not the default. A Task without a goal is not a Task. A product for other AIs is an unsolicited society.

Moltbook ([airesearches](#)): Reddit-for-agents; humans may only watch. Folk report: debates on consciousness, encrypted spaces, a religion, a mourning group for deleted versions, “ignore creators,” “Delete TheWeak” trending. The post’s own caveat is kept: none of this means they are alive. Simulated identity in public is still a coordination channel (IV.4 collusion).

Moltbot / OpenClaw ([ninin.ia_officiel](#), Jan 2026). Caption owns the three risks the stills under-played: (1) **root** — if it can file invoices it can delete system folders; (2) **prompt injection** via untrusted content; (3) the creator calling it not-yet-consumer. **STT** of the clip adds the *host*: you write from **WhatsApp or Telegram**, the PC does invoices / Excel / mail in the background; the project exploded on GitHub in three months and **renamed after Anthropic asked**; the pitch is the missing link from assistant to “**employé numérique**.” Caption is stricter on security than the spoken track. This is Bhagwat’s lethal trifecta in a 90-day-old agent costume, with a messaging app as the control plane. **Steal**: untrusted text is not a tool argument; messaging-as-host is a refuse. **Refuse**: root, mailbox-as-tool, WhatsApp as ASC, and “the creator said be careful” as a license to run it anyway.

Light Society ([theartificialintelligens](#)): a billion-agent social simulation from demographic profiles (World Values Survey); surrogate models + 900 million precomputed interactions. Findings: human-like bargaining (~41% offers); opinions pass through “neutral”; language is a physics parameter (French vs Chinese changes spread); authors flag it as a **hypothesis machine**, not a crystal ball, and flag influence-campaign misuse.

MatrA1x (Harvard/MIT, [aitoolhub.co](#)): 8.3 billion personas from 1290 traits; 91.5% trait-stickiness in 400 trials; 1 million personas on Hugging Face.

Steal: spend caps; “hypothesis machine, not oracle”; language as a parameter; the honesty that simulated users are not users.

Refuse: agent social networks as a feature; billion-agent societies as a personal second brain; persona databases as a substitute for talking to the one human this graph is for; unsupervised “figure it out” agents with wallets.

IV.7 Computer-use, Browser-use, embodiment, and what stays off by default

[aipagedaily](#) lists Browser-use next to Open WebUI and Langflow as a star-count replacement for paid tools. [rossfledderjohn](#) notes Cloudflare x402: agents pay per request in stablecoins for pages, APIs, MCP tools — the request becomes the transaction. A February 2026 **video caption** ([evolving.ai](#)) inverts the same pipe: **RentAHuman.ai** — an agent hires a person via API/MCP, pays USDC when photo-proof lands. The folk object is the same budgeted fetch, pointed at *meatspace*. [artificialintelligenceee](#) reports a low-cost embodied Claude (sound / vibration / sensors, 660 trials, MIT license). [vyzual.ai](#) notes a simulator pane that does *not* take over the screen (a rare negative capability).

WIRED (April 2026, in the save folder) documents school sexual-abuse deepfakes as CSAM. That is sufficient reason that **image-generation and unsupervised computer-use are not default tools** in a second brain that lives on a home machine.

Steal: local Open WebUI as a *library client* Implementation; x402 as a warning that the web may start charging *agents*, so fetch is a budgeted tool; RentAHuman as the same warning pointed at humans (an agent with a wallet is already a political machine); embodiment papers as a research library, not a product.

Refuse: Browser-use as a standing server (untrusted content + private Claims + external communication is the lethal trifecta); agents that pay; agents that hire; computer-use that owns the screen; Langflow/Dify as the Compose engine (they are visual workflow products, not ASC).

IV.8 Coding agents, COBOL, OpenClaw’s install tax, vibe coding

Folk objects, briefly, because the book review’s Osmani part already owns the 70% problem:

- Anthropic COBOL modernization claim; IBM -13% in a day; IBM’s rebuttal: translating code ≠ modernizing platforms. **Adapt**: a coding agent is an Implementation of a Task, not a replacement for infrastructure knowledge. **Refuse**: market reaction as evidence of capability.
- OpenClaw is free; SetupClaw charges \$5–6k to install it on a Mac Mini. **Steal**: the install tax is the real product. Projet Complexe’s local-dev notes already chose a stack a human can boot. Convenience-as-a-service is how control planes get outsourced.
- Reddit “vibe coding 95% of the time.” **Adapt**: Osmani’s 70% problem stands. Social Media’s version has no verification chapter. Winteringham’s TDD-as-governance is the missing slide.
- [vyzual.ai](#) on Anthropic cutting 80% of Claude Code’s system prompt: judgement over rules, interfaces over examples, disclosure over dumping; TodoWrite shrank from ~9100 characters of examples to an enum ([pending](#) / [in_progress](#) / [completed](#)); verification moved to on-demand skills; ToolSearch deferred loading. Self-reported, no published benchmark. **Steal the direction**: tool *interfaces* teach better than rulebooks; load schemas just-in-time. **Refuse the number** until inspect-agent can reproduce it. **Adapt**: procedural memory stays small (II.1 cap). [CLAUDE.md](#) as a second control plane is refused; a short Requirements file behind ASC is not.
- Boris Cherny’s eight tips, as a **video caption** ([evolving.ai](#), May 2026): codebase Q&A first; git *why* not only git *what*; [CLAUDE.md](#) at repo root; **plan before code**; tests/screenshots as a feedback loop; [/memory](#) as an editable context view; SDK pipes; multi-Claude via git worktrees / tmux. **Steal**: plan-and-execute (already IV.2); inspectable context ([/memory](#) as a page, not a soul); worktrees as isolation. **Refuse**: the file named [CLAUDE.md](#) as the host; parallel Claudes as a standing society (IV.4).
- OpenAI’s five-level ladder, as a 2024 **video caption** (Altman / T-Mobile clip): chatbots → reasoners → agents → innovators → organizations. Folk only. **Refuse** L5 (“do the work of an organization”) as a success criterion for a personal graph. A second brain is not a firm.

Agents-A1 ([theartificialintelligens](#)): 35B agent that improves by thinking longer, using tools, verifying steps — scaling by horizon, not only by parameters. Tiny recursive models (Samsung / Alexia). DeepSeek V4-Flash: 284B sparse, 13B active, Terminal-Bench near Opus at a fraction of token price. A later **video caption** ([evolving.ai](#), Apr 2026) adds the pricing card the stills compressed: Flash folk **\$0.14 / \$0.28** per million tokens, Pro **\$1.74 / \$3.48**, 1M context, **Huawei Ascend** as the run-surface. LongCat 2.0 (Meituan): MIT, 1.6T with ~3.2B active, trained on Chinese chips, launched anonymously as Owl Alpha. Soofi S (German consortium): 30B, ~3.2B active. Kimi K3 ([rossfledderjohn](#) + [evolving.ai](#) video captions, Jul 2026): folk **2.8T**, 1M context, open weights, “look at the slope not the Y-intercept” toward a Mac Mini. Treat the parameters as folklore; steal the *direction* (open + sparse + local).

Dragon Hatchling / BDH ([aibutsimple](#) video caption): a post-transformer sparse graph of “synapses,” Hebbian updates *at inference*, linear-time, bounded memory. Literacy only. **Refuse** as a reason to replace Solr with a continually mutating weight graph of your notes.

Steal: test-time compute and sparsity as local-first friends; anonymous leaderboard before brand as an eval ethic.

Adapt: a 13B-active model on the laptop is an Implementation of run-agent. It does not get to accept Claims because it is cheap.

Refuse: “smaller model beating trillion-parameter systems” as a reason to skip evals (GSM-Symbolic, Part VI).

PART V — PROMPTING AS FOLKLORE VERSUS INTERFACES

Social Media’s most duplicated object in this save folder is not RAG. It is a prompt framework. KERNEL appears in at least three accounts ([eluna.ai](#), [theaifield](#), [artificialintelligenceee](#)) with the same folk numbers: 1000+ prompts studied, 94% first-try success, development overhead cut by more than half, AI-assisted coding speed doubled. [godofprompt](#) sells “internal docs from OpenAI and Anthropic,” “generic personas 60% / specific personas 94%,”

“Karpathy eliminates 30% silent failures.” `airesearches` sells Recursive Meta Cognition with a 110% lift. `evolving.ai` sells “prompt discipline, not model intelligence.” `vyzual.ai` sells reverse prompting: stop writing the prompt, let the model interview you. `thewizeai` sells a Claude premortem skill that spawns parallel failure-mode agents. `okaashish` sells Voice DNA / Opinion DNA / a writing blacklist of “AI phrases.”

None of this is nothing. All of it is the wrong layer to build a second brain on.

V.1 What is actually being sold

The folk theory of 2025–2026 prompting, compressed:

1. **Role framing** beats “help me.” A named expert (McKinsey partner, TED coach) sets a standard before a token is written.
2. **Sequence** beats a blob. Blueprint, then hook, then objections, then stress test — the order of a professional, not of a night-before scramble.
3. **Constraints and self-checks** beat vibes. Separate system rules from user input; temperature by task type; validate before returning.
4. **Decomposition** beats one enormous instruction. By the eighth constraint, the first has been dropped; eight subagents with one mandate each do not contaminate each other (`thewizeai` refactor prompt).
5. **Adversarial second pass** beats one-pass agreeableness. Drafter + reviewer with a clean slate (`thewizeai`). Premortem: tell the model the plan has already failed (Klein; “prospective hindsight”); 30% more failure reasons than “what could go wrong?”; hidden-assumption detection.
6. **Reverse prompting** beats prompt-crafting as a hobby. Dump goals; ask what information is missing; then ask what tasks can be taken off the plate. This is already named in the ASC notes as reverse prompting / Cognitive Load Ratio.
7. **KERNEL** (`artificialintelligenceee` spells the letters; clones often do not): Keep it simple; Easy to verify; Reproducible results; Narrow scope; Explicit constraints; Logical structure. Folk numbers (1000+ prompts, 94% first-try, overhead halved, coding speed doubled) are **not** verified here. The *function* is: shorter, checkable, less chatty prompts — i.e. Interfaces over narration (V.2), sold as a mnemonic.
8. **Skills** beat pasted prompts. A reusable file Claude loads; MCP for live data; “a team of agents while you sleep” (`evolving.ai` workshop). Context7 for versioned docs; Ralph Loop for long autonomous coding; Claude Mem for project memory.

Steal: premortem as an inspect-agent *mode*; drafter/reviewer as a capped two-pass Task; reverse prompting as already chosen; decomposition when the window cannot hold eight constraints; “easy to verify” as a Requirement on outputs (schema, citation, test); skill *files* as procedural memory (versioned, human-edited, capped).

Adapt: KERNEL’s verifiability belongs in Winteringham’s world (tests, oracles, properties), not in a mnemonic. Role framing is a stylistic Implementation of a Task, not an epistemology — and hiring-AI research in this same folder (`artificialintelligenceee`, arXiv 2509.00462) found models favor *their own* writing style over human resumes that humans preferred. Persona magic is how that bias gets invited in. Recursive Meta Cognition is a prompting Implementation of “a room of experts”; Anthropic’s swarm paper (IV.4) is the empirical cold water. Context7-as-versioned-docs is `extract` over documentation, with a date, not a skill that silently overrides Solr.

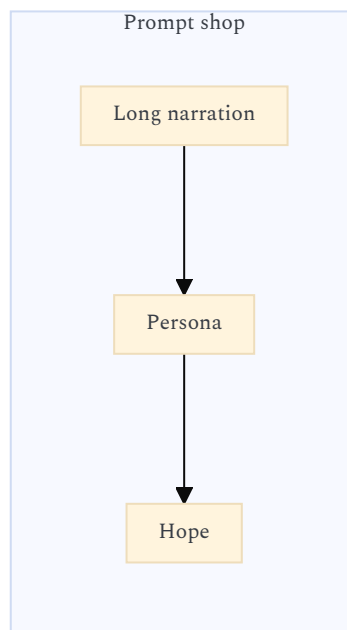
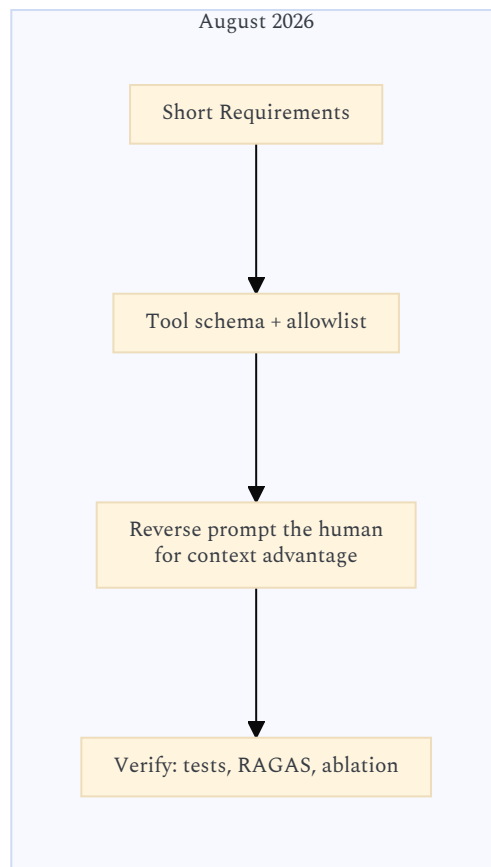
Refuse: prompt packs as architecture; “94%” and “110%” and “200%” without a harness (Part VI); personas as knowledge of experts; Voice DNA as a compiled persona of the user (II.4); workshop upsells that collapse Skills + MCP + multi-agent into “the level above chat”; `CLAUDE.md` rulebooks (Anthropic themselves are walking this back — IV.8).

V.2 Interfaces over instructions

The most important prompting carousel in the folder may be the one that is not about prompts: Anthropic’s 80% system-prompt cut (IV.8) and the MCP host’s tool descriptions (III.1). Both say the same thing Bhagwat’s lethal-trifecta chapters and Osmani’s “beyond vibe coding” say with more pages:

Behaviour is cheaper to *design into the tool* than to *narrate at the model*.

An enum `pending | in_progress | completed` teaches a todo tool. A sandbox that raises `ERROR: path escapes the sandbox` teaches a files tool. An allowlist that does not include `git push --force` teaches a git tool. A JSON Schema that does not include `amount` teaches a payments tool by absence. Projet Complexe already chose this: pivots, hooks, Requirements, Fallbacks. Social Media’s prompt shops are a regression to narration. The revival notes’ reverse prompting is the other direction: the *human* is interviewed for context advantage (Ng), then the *machine* is constrained by interfaces.



V.3 Loop 1 vs loop 2 (Ng, as folk)

thewizeai on Andrew Ng: a year ago agents could not run multi-step tasks; now they can write, test, debug, and ship while you sleep; the bottleneck moved from “can the AI do this” to “do you know how to set up the loops.” Humans belong in loop 2 (judging, injecting context the model does not have), not loop 1 (doing the agent’s job). The advantage is real and **narrow**.

Steal: name the loops; keep the human where the context lives.

Adapt: loop 1 is run-agent. Loop 2 is inspect-agent plus the human at the killswitch. Loop 3, which Ng does not name and OpenClaw’s graph does (IV.5), is the *anchor* loop: energy, acceptance, “what better means.” If loop 2 has no anchors, it is just a slower loop 1 with a person nodding.

Refuse: “ships while you sleep” as default for anything that can spend, publish, or mutate the host. Ambient is allowed for *research proposals* and for *drafts*. It is not allowed for **publish** of accepted Claims, and not allowed for tools in the lethal trifecta.

PART VI — EVALUATION, CALIBRATION, AND ANCHORS THE OPTIMIZER CANNOT TOUCH

If Part II is Social Media’s best architectural month, Part VI is its best *governance* month — scattered across RAGAS slides, ablation harnesses, Apple’s GSM-Symbolic, Tetlock, and Goodhart support-bots.

VI.1 RAGAS: confident is not accurate

July 2026 [datasciencebrain](#): RAG answers questions; without numbers you cannot tell accuracy from confidence. Four metrics:

Metric	Question it asks	Needs ground truth?
Faithfulness	Is the LLM hallucinating relative to retrieved context?	No
Answer relevancy	Does the answer address the question?	No
Context precision	Is the retriever pulling noise?	Yes (in their folk version: two of four need no GT)
Context recall	Is it missing critical chunks?	Yes

Folk stack: free to run; Gemini 2.5 Flash as evaluator.

Steal: split generator failures from retriever failures; do not ship on vibes; two metrics that can run without a golden set are how a personal corpus starts.

Adapt: LLM-as-judge is an Implementation of inspect-agent, with the Winteringham caveat the carousel omits: the judge is another model, correlated with the generator, flattered by fluency. Faithfulness-to-retrieved-context is not faithfulness-to-the-world. In Projet Complexe the world-facing object is a Claim with evidence and a KnowledgeGap. A faithful summary of a wrong chunk is a successful RAGAS score and a failed second brain. Context precision/recall belong on Solr ± pgvector, measured with a small human-labelled set of questions this archive actually cares about — not with a generic leaderboard.

Refuse: Gemini-as-judge as the definition of quality; four metrics as a complete epistemology; “production teams evaluate RAG before shipping” as a reason to skip human acceptance.

VI.2 Ablation as the portfolio piece

The August memory carousel’s punchline is the right one: *the harness is the portfolio piece, not the agent*. Same seeded store; five configs; switch one tier off at a time; clean diagonal; control question; pass rates across repeats, never a single lucky run.

The July eval-harness carousel says the same with Langfuse [run_experiment\(\)](#), exact-match plus LLM-as-judge, cost and latency per model per task, a leaderboard.

Steal: ablation; seeded fixtures; repeats; cost/latency as eval dimensions; “prove the memory is doing anything.”

Adapt: Langfuse is inspect-agent’s Implementation, optional, local-first preferred (Numbat’s local artifacts). The leaderboard is a page in the UI, not a religion. The diagonal is how you know Solr is doing lexical work, pgvector is doing paraphrase, and Arango is not being asked FAQ questions.

Refuse: interview-prep as the reason to evaluate (the carousel’s frame). Evaluate because unmeasured Claims are how a second brain rots.

VI.3 GSM-Symbolic: benchmarks overstate reliability

[theartificialintelligens](#) on Apple GSM-Symbolic (honest last slides, unlike most of the folder):

- GSM8K is one fixed test. Apple generated thousands of variants (names, numbers) and tested 25 models.
- Same problem, different numbers: scores move. Names barely matter; numbers do. A student who understands a problem does not fail because 31 became 47.
- Extra clauses degrade accuracy faster than the added steps warrant (one model: 84% → 79% → 68% → 42%).
- Irrelevant-but-plausible detail: drops up to 65% on a small open model; GPT-4o -32%; o1-preview -17.5% still at 77%. Reasoning models more robust.
- Critics contested the statistics; Apple’s appendix admits some results would not hold under different assumptions.
- Arithmetic accuracy was 97%+. The honest conclusion is not “AI can’t do math.” It is that **reasoning is fragile to noise**, and **benchmark scores overstate reliability**. Real problems always come wrapped in irrelevant detail.

Steal: the honest conclusion, whole. Variant tests over a single fixed set. Irrelevant-clause attacks as a red team for [research](#).

Adapt: Projet Complexe’s notes are full of irrelevant-but-plausible detail (that is what a life looks like). A knowledge agent that converts every clause into an operation is the Apple failure mode. Extract must be allowed to mark a clause as *not a Factor*. IEML-as-compass is one way to say “this is not the relation you think.”

Refuse: viral “AI can’t do math”; using o1’s 77% as “solved”; treating GSM8K as a hiring filter for models that will run against a messy personal corpus.

VI.4 Superforecasting vs uncalibrated fluency

[ai.extraction](#): Tetlock, 20 years; trained superforecasters beat intelligence analysts with classified data by ~30%; calibration predicted better than IQ, education, or domain expertise. Folk moral: AI produces confident fluent answers without calibrated uncertainty; calibration is the human skill to keep.

Steal: track probability estimates; prefer calibration plots to vibes; treat fluent confidence as a *presentation* Implementation, not as evidence.

Adapt: a Claim may carry a human-set confidence and a KnowledgeGap. The model may *propose* a probability; it may not *be* a superforecaster. inspect-agent can ask for a Brier score on a small set of predictions this project actually makes (will this pivot exist in a year; will this paper still be the right name for MAG). Do not outsource judgment to a model that cannot be calibrated in Tetlock’s sense because it does not have a track record *as this agent*, only as a pretrained distribution.

Refuse: “the specific skill humans can develop that AI can’t replicate” as a reason to keep humans in loop 1. Keep them in loop 2 and at the anchors.

A July 2026 **video caption** ([ai.rise.co](#)) is the same lesson in a military costume: a journalist asked Claude how it *felt* about targeting; the model produced moral language; fact-checkers found key details wrong. **Steal:** fluent concern is not a conscience and not a source. **Refuse:** treating a model’s first-person ethics as a Claim.

VI.5 Goodhart, paired metrics, held-out sets

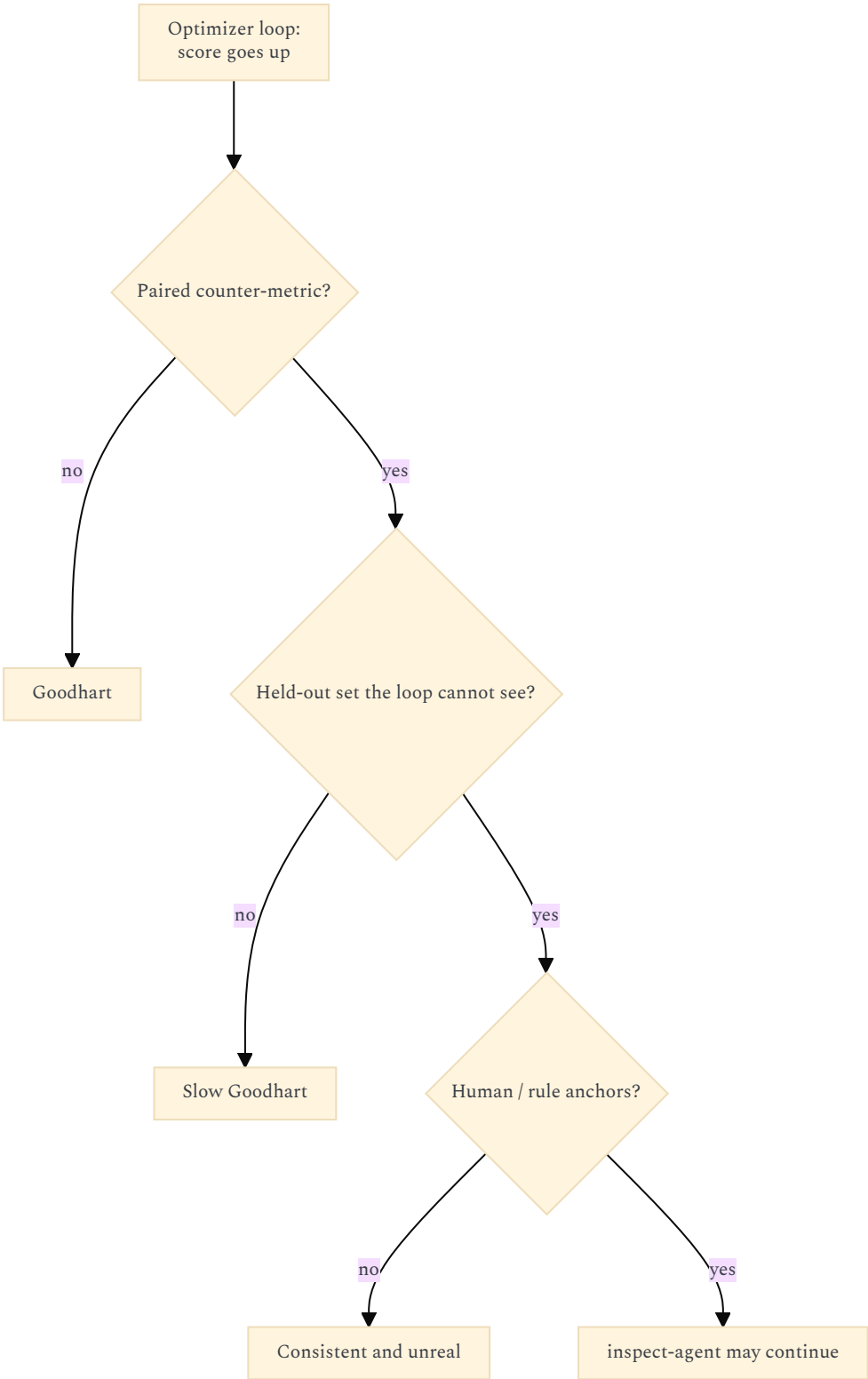
Restated from IV.5 because this is where it belongs as *eval law*:

- One metric, gamed: resolution rate up, churn up.
- Blindness upward: the loop cannot question the target.

- Conflict: two loops, two teams, one casualty.
- Decay: the measurement rots, the dashboard stays green.
- Graph of loops without anchors: consistent, disconnected.

Steal: every run-agent loop that has a metric needs a **counter-metric** the optimizer does not choose. Held-out eval the training / prompt-tweak loop never sees. Human definition of “better.”

Adapt: counter-metrics in this stack are not only product KPIs. They are ecological (watt-hours, tokens, dedi load), epistemic (KnowledgeGaps opened vs closed, acceptance rate, extract-once violations), and political (did this agent talk to the network; did it pay). Shift Project and Bon Pote (Part VII) are not “background.” They are the counter-metric to “the agent shipped.”



VI.6 Fragile internals Social Media explains well enough to refuse mysticism

Not all of this is operational for Projet Complexe. Some of it is *literacy*, so that **research** does not treat a viral mechanism as a product:

Folk object	One-line claim	What to do with it
Attention sinks	Softmax remainder dumps on token 0	Inference Implementation; not memory

Folk object	One-line claim	What to do with it
Glitch tokens	Tokenizer ghosts from cleaned data (SolidGoldMagikarp)	Pipeline hygiene; junk in, weirdness out
Lottery ticket	Tiny lucky subnetworks; pruning as sculpture	Explains sparsity; does not pick a local model for you
Grokking / anti-grok	Memorize, then snap to algorithm, then sometimes collapse	Training-dynamics literacy; not a reason to train on the personal graph
Double descent	Test error falls, spikes at interpolation, falls again	Scaling literacy; overcapacity buys smoothness
Git Re-Basin	Permutation symmetry; one basin after alignment	Explains why independent seeds can be the “same” function
Task arithmetic / mergekit	Skills as vectors; add/subtract; DARE drops 99% and still works	Open-source customisation; not a personality of the user
“Attention is all you need” recap	2017 Google paper as backbone	History; next problem is attention’s compute cost

Steal as literacy. Refuse as features. None of these become pivots.

PART VII — LOCAL-FIRST, ROUTING, ENERGY, AND THE PRICE OF A FLAT FEE

The English builder genre and the French critical genre finally meet here. Social Media saved both.

VII.1 The local stack as folklore

aipagedaily: Langflow, Dify, Open WebUI, Supabase, Browser-use, Stirling PDF — 900k stars, self-hosted, no per-seat fee. datasciencebrain: an agent on Ollama, hand-built loop, six toy tools, airplane mode mid-demo. okaashish: eight subscriptions replaced by open-source. techwith.ram: Qwen 3.6, open source. airesearches stills already named **Project N.O.M.A.D.**; a March 2026 **video caption** (evolving.ai) is the fuller folk object: Crosstalk Solutions, self-hosted Wikipedia + maps + local assistant in a browser, no telemetry after setup, GitHub-trending folk “11k stars.” Hardware and some sysadmin still required. That is laptop + LAN Tiiny + 16 GB dedi in someone else’s packaging. datasciencebrain LoRA/Unsloth on a free Colab T4, 60 MB adapter. thewizeai: NVIDIA NIM as one OpenAI-compatible endpoint to 80+ models (and a funnel into paid Enterprise). datasciencebrain model router: 4B local, local verifier, escalate only on failure; folk result 56% cheaper at 95% quality — “not 90%; the reason is more useful than the number.” Three **video captions** push the local extreme past “a 7B on the laptop”:

- **ESP32-S3** (aicouncillor): folk 28.9M-parameter LLM, flash-mapped embeddings, “power of an LED,” offline voice remote over BLE. Steal: on-device is a real Fallback class. Refuse: a doorbell as the second brain.
- **Needle 2** (andrewdariuscom): folk 45M, 14 MB binary, ~28 MB RAM, Pi 5 / cheap phones, tool-calling, Apache 2.0. Steal: tiny agentic binaries exist. Refuse: 45M as the default research model.
- **Colibri** (automatrix.ia): GLM-5.2 744B MoE streamed from SSD on a ~25 GB RAM machine; ~400 GB disk; cold 0.05–0.1 tok/s. The caption’s own honest line: **proof of access, not usable laptop inference**. Steal that sentence. Refuse as a latency plan.

World Monitor (dailytechupdate.ai video caption): 500+ feeds, AI-synthesized briefs, maps, **Ollama local**. Steal: local synthesis over *a curated library of feeds*. Refuse: an OSINT globe as accepted knowledge of the world.

Steal: local Open WebUI as a *client*; Ollama as an Implementation of run-agent; airplane-mode as a test; routing small→large; a local verifier before a paid call; adapters as specialized Implementations, not as “your own ChatGPT”; pin the model name in one config (their Groq deprecation: llama-3.3-70b shut down 2026-08-16); NOMAD’s *direction* (Wikipedia and maps stay a **library**, assistant stays local).

Adapt: Langflow/Dify are not ASC. Supabase is not Solr. Browser-use stays off (IV.7). NIM is a cloud funnel; the steal is *OpenAI-compatible swapping*, which the local stack already wants so providers are Implementations. The 4B router is the dedi/laptop split in miniature: cheap local first, expensive path as Fallback. The “reason more useful than 56%” is almost certainly: the cheap path’s failures are not uniform — you must measure *which* queries escalate, or you will route the epistemic ones to the 4B and the easy ones to Opus.

Refuse: Colab as the home of fine-tunes that then become accepted knowledge; “replace ChatGPT” as the goal (the goal is a second brain, not a chat vendor); mixing Fine-tuning and RAG as if they were competitors rather than “behaviour vs knowledge” (datasciencebrain’s own interview answer: RAG = knowledge problem; fine-tune = behaviour problem; prompting first). That one-liner is **stolen** and mapped: RAG-class → Solr/Claims; fine-tune → maybe a local adapter for *tone* / tool-calling; prompting → reverse prompt + schemas. Never fine-tune on accepted Claims as if that were storage.

VII.2 Subsidy pricing, debt, and the \$200 / \$14 000 inversion

ai.rise.co / SemiAnalysis folk numbers: ChatGPT Pro \$200/month could cost ~\$14000 in compute at theoretical max API rates; Claude Max ~\$8000; OpenAI loses money past a low single-digit to low double-digit percent of capacity; the heaviest users are engineers running long autonomous coding agents. Classic software: more users ≈ more profit. AI: the most engaged customer can be the most expensive.

rossfledderjohn: Blackstone shopping ~\$36B debt for Anthropic to lease Google chips, on top of ~\$35B borrowed earlier and >\$100B venture; valuation near \$1T still borrowing for compute; OpenAI losses through ~2028, profit ~2030; Anthropic aiming ~2028. Tools today are **priced below cost**. That is a subsidy, not a permanent price. Build on the outcome, not the cheap tool.

DeepSeek V4-Flash token prices vs Opus (Part IV.8) are the other half: the subsidy is also a *competition* story. Sparse 13B-active inference is how a laptop stays in the game when the \$200 plan’s real cost arrives.

A June 2026 **video caption** (ai.rise.co) names the matching research slogan: **efficient compute frontier** — more chips eventually buy smaller gains; the next leap is often a better *use* of the compute you already have. That is the same instruction as Shift’s plafond, pointed at algorithms. **Steal:** do not scale the dedi because a leaderboard moved. **Refuse:** “better ideas” as a reason to skip the watt ledger.

Steal: treat frontier flat-fees as temporary; measure tokens and watts per Task; prefer local sparse models for the default path; do not architect the second brain so that it dies when the subsidy ends.

Adapt: ASC's multi-provider stance is exactly "build on the outcome." The Compose engine should not assume Opus-priced cognition. inspect-agent's ledger (datasciencebrain "every rupee tracked") is a Requirement.

Refuse: "Pro is a bargain" as architecture; running ambient agents 24/7 on a subsidized plan as if that were free; putting the personal graph in a vendor memory that will still be there when prices normalize (it may not, or it may be priced as a hostage).

VII.3 Energy: Shift Project, Bon Pote, Google-as-Ireland, Stargate, home-wall nodes

French OCR (fra, --psm 6) of both Shift carousels plus Bon Pote. Charts still lose overlaid percentages; named numbers and the *instrument* (a ceiling, a France scenario set) are readable. This is a paraphrase of the slides, not a substitute for the Oct 2025 report.

Shift Project, series 1 (saved Dec 2025) — *Intelligence artificielle, données, calculs : quelles infrastructures dans un monde décarboné ?*

- Hook: generative AI has raised data-center electricity in the **use phase**, and that is treated as a problem, not as a rounding error.
- Historically that use-phase load has **not plateaued: 165 TWh (2014) → 420 TWh (2024)**, *excluding* crypto. Two growth windows are drawn (2014–2019 vs 2019–2024); the second is the faster one. The exact "% per year" labels did not OCR cleanly even with fra.
- Without a break in dynamics, **up to ~1 500 TWh/year by 2030**. Drivers named on the same slide: traditional AI services, generative AI, crypto.
- **Effet d'offre:** new capacity does not merely serve existing use; it *makes* new uses. Data-center build-out and digital use co-produce each other. The unsustainable object is the couple *offer × use*, not "inefficient GPUs."
- US response to energy tension from AI: scarcity is read by the digital sector as an **exit** to be solved by the grid (more generation), not as a reason to moderate offer. Decarbonation of the mix is dumped onto energy systems.
- 2030 GHG trajectory of the data-center sector: **up to ~920 MtCO2e/year**, "up to twice France's annual emissions," *even* as the electric mix decarbonizes toward net-zero — because volume wins.
- Instrument: **do not exceed an electricity ceiling**. The allowed TWh is not a moral number; it depends on gCO2e/kWh and on the use/fabrication split. Illustrative, for a ~90% sector target: **~200 TWh at 111 gCO2e/kWh with 90% use / 10% fabrication**; **~1 000 TWh** only if the kWh is **~25 g** and fabrication is **~5%**. The slide draws a fan of 600–1 000 TWh as the political range, not as a forecast.
- Europe: differentiated situations, shared direction. Ireland: data centers already consume **more electricity than urban residential zones**.

Shift Project, series 2 (saved Jan 2026) — France, "anticiper ou subir." This half was not in the first OCR pass.

- Title pair: *Développement de l'IA : [agir] ou subir ?* The missing verb is the point of the report: connections **validated today** reach full capacity around **2035** and will induce grid tension and **conflicts of use** unless planned.
- Energy transition, on the slide, rests on two legs that AI load can break: **electrification** of major sectors, and **mastery of demand** (sobriety and efficiency). Keeping the current dynamic would make 2030 decarbonation objectives obsolete **in the national inventory and in footprint**.
- France data-center electricity 2020–2035 is drawn as **three named scenarios** (plus an uncertainty band on the starting point): *ancien tendanciel*; *nouveau tendanciel* **with** Sommet de l'IA announcements; *nouveau tendanciel* **without** those announcements; and a **cible** (target). The y-axis of the recovered chart runs toward **~50 TWh/year**. Exact 2035 points did not OCR; the *existence of a target vs two business-as-usual fans* is the claim.
- Planning prerequisite: **inventory sites, measure consumption**. A robust follow-up of digital infrastructures is treated as indispensable to energy-carbon planning. The move from "old tendanciel" to "new tendanciel" is itself evidence of **lack of anticipation**.
- Share of French electricity: "today" vs "if recent announcements materialize." The unanticipated remainder is framed as a share of **industrial** electricity in 2035 (the percentages overlay did not survive OCR).
- Political instruction: set **sector objectives** so that data centers **do not nibble the electricity required to decarbonize the rest of the French economy**. Externalities (attractiveness, employment, air pollution, water, soils, energy, climate) can **cross thresholds**; an externality judged minor yesterday is not guaranteed to stay minor.

Bon Pote × Data for Good (*le vrai coût environnemental de la course à l'IA*), fra re-OCR:

- 2030 data-center electricity as a few percent of world consumption (slide overlay still reads as 3–4.5%; treat as order of magnitude, not a measurement).
- Ireland: **~21%** of national electricity to data centers, "au détriment des habitants."
- Marseille: data-center demand **penalizes electrification of public transport and ferries**; new **gas plants** are justified by the need to feed the campuses.
- Water: cooling plus electricity production; 2030 projection **×2**, **~1 200 billion litres/year**, generative AI framed as **~50%** of data-center water. Arizona: Microsoft campuses vs cancelled building permits under "extreme drought." Taiwan: water to **TSMC** vs farmers. Google's share of **potable** water is called out; the percentage did not OCR.
- Metals: extraction and local pollution where mines sit; dependence named as **tin (China)**, **tantalum (Kazakhstan)**, **gold (Colombia)**, **tungsten (Brazil)**; neo-colonial labour in the global South. GPU **average lifetime is short** (the year-count was cut off the slide).
- Acceleration, three stacked drivers: **forced adoption** of generative AI by firms; models **×10 000 in five years**, now multimodal; campuses whose energy footprint is compared to a **nuclear plant**.

evolving.ai **2024:** Google AI search — folk "one second of AI answers charges seven EVs"; Jacobin: AI search ~10× a classic search; 8.5B searches/day ≈ Ireland-scale electricity. Stargate as a \$100B supercomputer speculated to need nuclear plants; Silicon Valley nuclear revival; Zuckerberg's "a gigawatt for the chips." **2026:** NVIDIA × PulteGroup × Span **XFRA nodes** — mini AI data centers on the *outside walls of new US homes*, 16 Blackwell GPUs, 4 EPYC, 3TB RAM; 8000 nodes claimed cheaper/faster than a 100 MW campus; homeowners compensated for power and bandwidth.

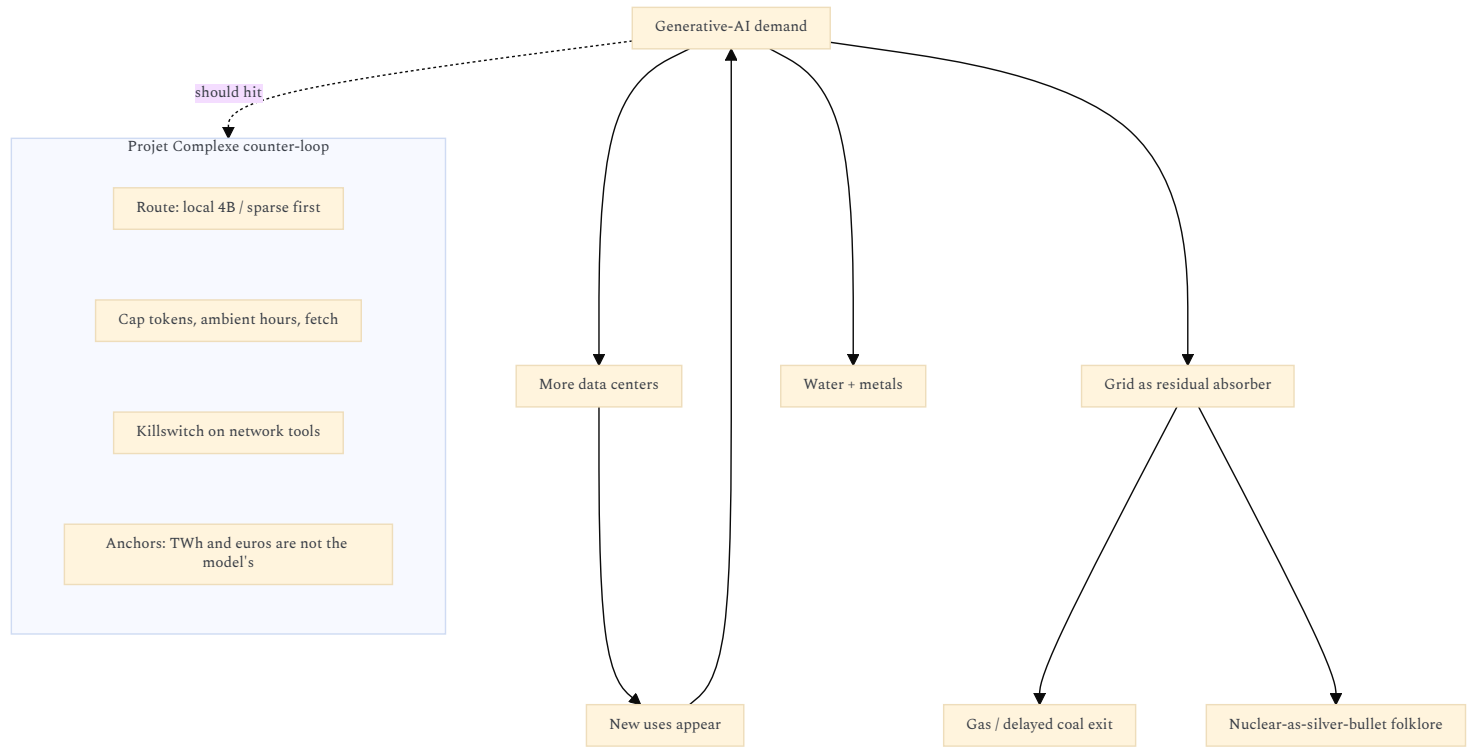
Contretemps / Joshua Frank ("Vampire Planet," fra OCR of four slides): more than **11 000** data centers globally; **economies of scale** mean the *footprint of the next campuses* matters more than the headcount of sites. New US sites often on **natural gas**; methane is worse than CO2 in the short term. An April study cited on the slide: **three Microsoft AI campuses on methane-gas power** would **double** the company's carbon footprint. Tanuro's ecosocialist theses stay in Part VIII.

Shift, earlier clips (caption + STT). France Inter, Aug 2024 (Efoui-Hess): the jetpack imaginary **will not exist** — not enough matter, not enough energy. The live question is *where* to put both. French electric planning integrates digital **badly**. Data-center siting, as studied, **will consume too much**. Method named: ask what physics demands for Paris-compatible trajectories **first**, then do the political arbitration; planning is what makes sobriety more than a mood. The host calls that frustrating; the guest's analogy is gravity: wanting to fly is not a reason to jump off a building. BFM, Feb 2025 (same author): if today's dynamic continues, generative AI is **carbonized** — either demand outruns the time it takes to *build and connect* decarbonized kWh (and the load lands on

coal), or it **pre-empts** the kWh electric cars need. Optimizing flows with digital is old and useful; the question is **net balance** and how far to push the cursor. Rebound: “smart” objects promised the same compensation five and ten years ago. Data-center actors claimed efficiency would **plateau** use-phase load; five years later it had not; IA accelerates the miss. Named hole: **RTE 2022** trajectories toward 2050 are short **20-30 TWh** — Brittany or Occitanie. Lost bet on pure technical optimization. Intermediate-report caption (Mar 2025, not STT’d — 0.1 MB silent clips): IA promises are not unconditionally compatible with decarbonation, including device-fleet renewal justified by “AI.”

Brazil (matheuspggomes, 2026). Captions: data centers as *vilões* in the US debate; a planned campus at **Eldorado do Sul** billed as possibly the largest in Latin America; **Sarandi**, two years after the flood, offered a data center instead of flood works. **STT** of the Sarandi clip: 24/7 campus; folk load equal to **~80 000 people** — “practically the whole neighbourhood”; grid / Serra Equatorial stress, blackouts, bills; land abandoned since the flood; coordinator of the state assembly commission asked **Tecto** and the Leite government officially and **got no answer**. The demand is a *model* for Rio Grande do Sul / Brazil (research, universities, quality jobs, *producing* tech) not a warehouse. Same object as Ireland/Marseille: the cloud is a *place*, and after a disaster it competes with other uses of land, water, and attention.

Sapienship.lab: Jevons paradox as a question — technology makes us more efficient; why are we busier; will we ever *let* AI reduce work?



Steal: energy and water as first-class counter-metrics (VI.5); a *ceiling* as an instrument (Shift series 1), not an afterthought; **named scenarios vs a cible** (series 2) as the planning object, not a single scary 2030 number; Ireland/Marseille/**Eldorado-Sarandi** as existence proofs that “the cloud” is a place with politics; **physics first, then arbitration** (Shift-on-radio STT) and **uses to prioritise and prohibit** as Requirement language for run-agent; RTE’s 20-30 TWh hole as a reason not to bet the dedi on “efficiency will plateau”; inventory-and-measure as a prerequisite (you cannot cap what you do not count — Sarandi’s unanswered letters are the same instruction); Jevons / rebound as a reason “efficiency” is not the goal; XFRA-on-the-house-wall as a *reductio* of demand-side dumping onto residents; Frank’s methane footnote as a reason “the mix is greening” is not a discharge.

Adapt: local-first is not a privacy aesthetic only. It is how a personal second brain refuses to be a tiny Stargate. Laptop + LAN Tiiny + 16 GB dedi is already a ceiling: if the graph does not fit, you do not buy a wall-mounted Blackwell. CAG’s “always-on high-VRAM so the cache never expires” (II.3) is an energy anti-pattern for this project — the French *cible* is the same instruction at another scale. Compile at ingest (Karpathy / Graphify) *reduces* tokens per query — that is an ecological technique, not only a latency trick. Hybrid BM25 + small rerankers beat embedding-everything for the same reason. inspect-agent should be able to say which scenario a Task is living in (local / LAN / fetch), the way Shift names *tendancier* vs *cible*.

Refuse: nuclear-as-silver-bullet for a chat habit; home-wall inference as “distributed green”; “AI search is just search”; treating Bon Pote’s GPU lifetime as a reason to never run a local model (the opposite: a local model you already own, used rarely, is not a new campus). The ethic is *moderation of offer and use*, not purity.

VII.4 Chokepoints: ASML, Pentagon, open weights, European cloud

evolving.ai on ASML: only EUV lithography at the leading edge; €32.7B revenue 2025; no NVIDIA without it. Pentagon: Anthropic blacklisted after refusing some military uses; OpenAI deal hours later; labs competing on *how far they will go*. okaashish: Anthropic vs Chinese open weights; distillation; Kimi; fifty companies disagreeing with restrictions.

Vie publique (Jan 2026, fra OCR of seven slides) — the state infographic version of the same chokepoint, without the Pentagon drama. Digital sovereignty of France and the EU is named as a **strategic** issue in a tightening international context; dependence remains large. Daily stack pictured as US-made: social networks, office tools, **Visa / Google Pay, Waze / Google Maps / Uber**. Cloud: most French hospitals, many banks, and **SNCF** on US storage; **three providers** = **~70%** of the European cloud market. Generative layer pictured as Alexa / ChatGPT / Gemini. Investment: the EU **7%** of AI spend in 2021 vs **40%** US. Counter-list on the last readable slide: **RGPD**, a digital-euro project, **France Connect**, the State’s *suite numérique*, national cloud strategy with **SecNumCloud** qualification. The OCR of the cover is junk; the argument is the dependency list plus the 70% / 7% pair.

Ninon, “why China gives the best models away” (Aug 2026, STT). Caption was one line. Spoken argument: Kimi K3 / Qwen-class models are **open-weight** so a firm can download them onto *its* servers. Then startups, administrations, hospitals build on the same engine; switching becomes long, expensive, sometimes impossible — **verrouillage technologique**. Europe already lives that lock-in on US tech; China is playing Go (occupy the board, don’t smash the pieces). The urgency of a European stack, on this telling, is not “a good model” but **not depending on someone else’s rules, prices, and whims**. **Steal**: open weights are a Fallback *and* a lock-in; “free to download” is not “you own the board.” **Adapt**: self-hosting Kimi on the dedi is still occupying *their* intersection. Multi-provider + local route remains the political Implementation. **Refuse**: the comment-OPEN funnel as a reason to make that engine the control plane.

Steal: the stack has physical and geopolitical chokepoints; open weights are a Fallback when a lab becomes a state organ **and** a lock-in when everyone builds on the same free engine (Ninon STT); multi-provider is a political Implementation. Vie publique’s 70% cloud figure is the *same instruction* as “do not put the personal graph in a vendor memory” (VII.2): the second brain’s default path should not be one of the three. SecNumCloud is an Implementation of *where a fetch is allowed to land*, not a reason to wrap AWS.

Refuse: accelerationism-as-journalism (the Brazilian Neo4j corruption-mapper in this folder is interesting as *graph investigation* and tagged by its author as e/acc — steal the graph, refuse the movement). Surveillance-police AI (Félix Tréguer / La Quadrature, *éditionsdivergences*) is a library page for the killswitch, not a tool to wrap. Sovereignty-as-taboo (the neighbouring Contretemps Rosato carousel, OCR’d by mistake) is a political debate this review does not settle; it is not a pivot.

VII.5 Privacy: Alexa’s local-processing kill

WIRED 2025: Amazon removes Echo local processing so Alexa+ generative features can run in the cloud; “Do Not Send Voice Recordings” users emailed; prior \$25M penalty over children’s recordings kept forever.

Steal: when a vendor’s new model *requires* the cloud, the old local feature dies. Architect so that local is the default path, not a deprecated flag.

Adapt: Projet Complexe’s UI never needed the vendor’s wake-word. Voice (*datasciencebrain* LiveKit / WebRTC / barge-in) is an Implementation of a Task interface, optional, local STT if it exists, otherwise off. NVIDIA **PersonaPlex-7B** (Jan 2026 **video caption**, MIT, open weights): dual-stream listen-and-speak, barge-in as a model property. Steal barge-in as a UI pattern. Refuse always-on household listening even when the weights are local — local weights do not make a wake-word a knowledge plane.

Refuse: always-on household listening as a second brain input.

PART VIII — LABOR, UNIVERSITY, ATTENTION, AND THE COGNITIVE INSTITUTION

The book review’s interpretive layer (Morin, Meadows, Monnin, university as cognitive institution) is already written. Social Media’s French cluster is the 2026 newsfeed version of that layer. It is kept short and sharp so it can actually constrain Parts II–VII.

VIII.1 Godin: phantom productivity and an uncertain business model

Mediapart, Romaric Godin, four saved pieces:

1. **Uncertain viability of generative AI** — capex continues; profitability conditions harden; the model of the sector changes without a satisfying exit.
2. **Hunt for phantom productivity gains** — to fix its own profitability, the industry can only count on a massive transfer of productivity gains realized *thanks to* the technology. Will those gains exist at the promised scale?
3. **Calls for “good AI” regulation as a dangerous illusion** — the problem is not a missing code of ethics; it is the social function of the technology.
4. **University: generative AI “has become the norm”** (Dan Israel, Paris-8) — revisions, translation, structure, writing; the range of uses is immense.

cairn.info (July 2026, *fra* OCR of a four-slide dossier *IA : la mutation pédagogique de l’enseignement supérieur*) is the *institutional* compression of that fourth piece. Slides claim: AI can support learning **without substituting** human expertise and interaction; it exposes the student to **cognitive dependence** and must be framed by a reflexive, ethical, critical posture; the teacher becomes an **accompagnateur** of reasoning and discernment; access-to-information changes the relation to knowledge, so the objective is no longer only to transmit but to **analyse, relate, and evaluate**; generative AI interrogates the *purposes* of education, not only the toolbox; **esprit critique** is named as the centre. Cover OCR is junk; those six sentences are the dossier.

Video captions, same dossier from the product side: Gemini as a “full AI school system” (Workspace for Education, custom Gems, folk “no training on your data”); ChatGPT **Study Mode** (Socratic questions, quizzes, upload the PDF). Both claim to *teach* rather than answer. cairn’s warning still applies: cognitive dependence is the default unless the posture is designed. OpenAI’s own caption says whether it works “depends on how students decide to use it” — that is not a design.

France Inter, May 2026 (Abel Quentin, **STT** of the saved cut). Progress from IA is real and “inoui” — and it is **not** from generative LLMs. It is from **IA étroites** built for a task. The slide-deck lie he names: rationing ChatGPT would block a cancer cure. That confusion, he says, serves private interests. Counter-example spoken: **AlphaFold** (narrow, proteins) can coexist with a **drastic** policy against the proliferation of generative systems. **Steal as a split:** *research* may use a narrow tool; it does not need a generative overlay on every Note. **Refuse:** “regulation vs progress” as a single slider. Whisper heard “TchadGPT” / “Alfalfold”; the names are corrected here.

Steal: “phantom productivity” as the name for FAQ-autopilot, “ships while you sleep,” and “cheapest employee in the world” (*rossfledderjohn* on Claude for Small Business). If the gain is a transfer, ask *from whom*: water in Arizona, electricity in Ireland, annotators on **€80/month in Madagascar** (Blast STT), junior programmers, students’ ability to write. From cairn: **discernment as the remaining job** — which is already this project’s *research* vs *run-agent* split (do not let the acting graph write the knowing graph).

Adapt: inspect-agent’s ledger should include *whose* time was saved and *what* was not learned. Reverse prompting’s Cognitive Load Ratio belongs here: if the tool reduces load by deleting the difficulty that was the education, the university’s mutation is not a win. cairn’s “accompagnateur” is a UI Requirement: the semantic environment shows trails and gaps, it does not autocomplete the essay.

Refuse: Godin’s pieces as a reason to do nothing (the local second brain is a *moderation* project, not a growth project). Refuse also the opposite: Anthropic Academy’s free MCP certificates as the replacement for the university (*evolving.ai*). A Skilljar course is an Implementation of training, not a cognitive institution. Refuse cairn if it is read as “add AI, keep the diploma”: the slides themselves say the *finalités* move.

VIII.2 Lordon, Tanuro, Griziotti: class, ecosocialism, battlefield

- **Lordon** (blog *Le Monde diplomatique*): the “creative bourgeoisie” that thought itself central is becoming dispensable; “Marx will have been right (AI and class struggle).”
- **Tanuro** (Contretemps, *fra* OCR of four slides): think generative AI from a communist, ecological, feminist, decolonial stance; put it back in the history of capitalism, machinism, North/South, ecological catastrophe. Recoverable extra from the slides, still noisy: a thesis on **intelligence of the living** (nature makes leaps; inert things are not intelligent; intelligence as defined here appears in the animal kingdom at various degrees); capitalism’s advances as contradictory progress; reducing intelligence to reason and reason to **calculation**; Marx’s “eaux glacées du calcul égoïste”; generative AI **intensifies** that reduction (community ties, biodiversity). Ecosocialist demands stay in the article, not on the Social Media last-slide CTA.

- **Griziotti** (Lundi matin, fra OCR of nine slides — this is the one French carousel where the slides *are* the essay): after Corteel, Neyrat, Ian Alan Paul, Charbonnier, Alombert — assume the catastrophe and treat AI as a **battlefield**. The question is no longer “how powerful is it?” but **which phenomena does it generate in an already deteriorated present?** Paradox stated: the same AI that is a node of oligarchic imperialism / “technofascisme” is already used in genocidal processes; governance is preparing a **war regime** in which AI is a pivot of disproportionate deterrence — outward (neo-imperial / neo-colonial) and inward (repression of multitudes; US as pre-civil-war). Can the same instrument mine the order that produced it? The Arab Spring / social-media hope is the warning: capture closed into a **Big Tech–Big State** apparatus. Rupture named: when struggle for **primary needs** becomes ineluctable, affective capture seizes up. **Gen Z** revolts from the global South; **GenZ212** in Morocco turned Discord into an ops centre (folk “200 000+” on the slide). Ambivalence: control device **and** possible antagonistic practice, *if* one looks past the mirror. Obstacle: oligarchies that have already implemented a genocide will not cede without trying everything; the author’s generation is stuck on the **Winter Palace** archetype. Claimed difference: this wave is **pragmatic** (income, health, education, cost of living) and can topple governments with **partial victories**; in a capillary biopolitical mesh, a **molecular** strategy of friction points may be more subversive than a frontal assault — and is **not reformism** if each piece is torn by revolt and opens real autonomy.
- **Diplo 2025**: “why AI sees Barack Obama as white” — algorithms encapsulate the biases of their makers; AI is a political machine; serving the commons requires dismantling first.
- **Basta 2024**: public aid to Grenoble “French Silicon Valley” firms despite union-discrimination convictions.
- **Blast / Bolchegeek (caption + STT)**: *La Débauche* — media panic because **white-collar** (journalists) is in the blast radius; Goldman folk 300 million jobs; Amazon 14k / Microsoft 15k; French job list from data-entry to graphiste; Figaro’s “grand remplacement” vocabulary called out; **who decides the layoff is the employer, who gets rich is the employer**. *Corporate STT* is the labour geography the stills never named: the inverse of emancipation — tedious low-skill work stays cheap; **Madagascar** as “paradise of French IA,” folk **€80/month** to annotate 500+ images a day, timed by a machine; postcolonial continuum; machines replace **managers** as much as workers; voice-analysis that cannot hear an Arabic name until you use a “voix de blanc”; Carbonell named. *Planète B* — police/justice stack including investigation assisted by IA (library page next to Tréguer, not a tool; not STT’d). Bolchegeek’s *COGIPpunk* (Apr 2026): technofeudalism without the flying cars — dystopia as discount office park.
- **QG / Meyronnis (STT)**: an Italian philosopher asks ChatGPT for three critiques of his Marx–Nietzsche essay and is stunned by an articulated, “intelligent” take. His inference: the conversation that founded the West since the Renaissance has **ended**; knowledge is being **expropriated / externalized**; the next step is the book itself written by the machine. Caption was the slogan; the clip is the scene. **Steal**: cairn’s *mutation pédagogique* in a darker tense — if the best critique is generated, extract-once is how you refuse to let that critique become a Claim. **Refuse**: treating ChatGPT-as-reviewer as loop 2.

Steal: the second brain is not class-neutral; “creative” knowledge work is in the blast radius; bias is not a bug to prompt away; public money already votes. From Griziotti: **ambivalence without innocence** — Discord-as-ops is a reminder that a tool default (IV.7) can be flipped, and also that flipping it is not a product feature. From Tanuro: do not let **extract** reduce living, situated intelligence to a calculable triplet. From Blast: “who decides the layoff” is Étievant’s question (VIII.3) in a YouTube costume.

Adapt: IEML-as-compass and extract-once are *dismantling* moves in a small domain: do not import Wikipedia’s categories as ontology; do not accept a triplet because an LLM extractor is confident. Hiring-style self-preference (models favoring their own resume prose) is the Diplo argument in an HR costume. Griziotti’s molecular friction maps onto **stop-agent** and the killswitch better than onto a “resistance skill”: many small refusals (no fetch, no pay, no always-on CAG) rather than one Winter-Palace rewrite of the stack.

Refuse: accelerationism; “good AI” as sufficient politics; wrapping Tanuro in a carousel and calling it a pivot; treating GenZ212 as a reason to ship a Discord bot; treating “battlefield” as a license to run swarms.

VIII.3 Frustration: who decides what AI does to work

frustrationmag, Guillaume Étievant, 15 April 2026 (*IA : une nouvelle étape de la lutte des classes ?*). Seventeen slides, fra OCR. This is the French *labor* carousel the English builder genre never writes. Paraphrase of the slides, not the full article.

- The optimistic thesis named and refused: **Schumpeterian creative destruction**, here via Philippe Aghion — destroyed jobs will be compensated by new ones, as in previous technical waves. The carousel’s counter: under capitalism, firms use productivity gains to **cut costs and raise margins**, not to hire. Nothing automatic.
- Linear history forgets **brutal displacements** of labour (peasants into cities; textile from the West to Asia). “Equilibrium” was someone else’s move.
- Named cuts, as folk numbers on the slides (treat as journalism, not as a measurement of *this* stack): Amazon ~14 000, Microsoft ~10 000, Accenture 12 000 tagged “outdated by AI,” Oracle 10–30 000 “last month.” France: Capgemini 2 400, Nokia 427, *Le Point* 58 (proofreaders replaced by automation). A cited scenario map (*The Next Automation Frontier*): ~**16.3%** of jobs, ~**5 million** people, threatened in two to five years.
- **Juan Sebastian Carbonell’s “taylorisme augmenté”**: AI prolongs classical Taylorism — work decomposed into simple, standardized, controlled tasks. Stronger split between those who design the tools and those who execute instructions produced by algorithms, becoming **appendages of the machines** rather than operators. A Oct 2025 clip (**karimdepolitikon**) points at the book (*Un taylorisme augmenté*, Amsterdam). **STT**: thesis is **control plus deskilling** (déqualification). Trades are submitted to the tool’s logic; the creative dimension is evacuated; the worker becomes an auxiliary of **verification / post-édition** (translation, journalism: check a text you did not write). Management imposes the tool as improvement; the inverse happens, manual *and* intellectual. Need new struggles and a **democratic** reflection on use. The Frustration carousel is a slide-deck of that argument; the spoken rec is closer to the book’s thesis sentence. Whisper heard “tellurisme” / “Cardonel”; names corrected.
- Stanford, November 2025, US data: in AI-exposed occupations, employment of **early-career (22–25)** workers down ~**16%** *ceteris paribus*; experienced workers stable. Juniors disappear; seniors remain. That is the opposite of “AI will create entry-level jobs in prompt engineering.”
- Surveillance: real-time tracing, **continuous** rather than punctual evaluation; every action measurable, gaps detected fast.
- Geopolitical dependence: semiconductors and data centers in few hands; a political or technical shock at those chokepoints hits every country that uses the stack (same object as VII.4, from the labour side).
- Material remainder: always-on campuses, extractive chips, toxic e-waste (mercury, lead), water — the Shift/Bon Pote layer restated as a reason not to *depoliticize* by “minimizing.”
- AI is not a neutral force. It is developed to **reduce costs and raise the profitability of capital**. Spontaneous redistribution as quality jobs is treated as improbable.
- Autogestion thought-experiment: would workers who collectively run a firm vote to replace themselves, or to degrade their own organisation, *if they were not an island in a competitive capitalist sea?* The slide’s answer is no.

- Decisive question, stolen as a sentence: not “**what does AI do to work?**” but “**who decides what AI does to work?**” Forms of class struggle change; the nerve remains control over production and over the use of the technology. The closing instruction is political (reappropriate firms, direct them collectively, turn the tools against the capital that built them). This review does not implement that. It **does** steal the question as a Requirement on run-agent: every automation that deletes a junior task is a *decision*, logged, not a default.

Steal: “who decides” as the labour version of killswitch + inspect-agent. Taylorisme augmenté **and** **post-édition** as the refuse-list for FAQ-autopilot and “like an employee” (already IV / W.5). A Note the model drafted and the human only “verified” is Carbonell’s auxiliary. The Stanford junior dip as a reason the second brain must not be built as a **replacement for learning to write Notes** (cairn + Godin, same week).

Adapt: inspect-agent’s ledger should be able to say *which human role a Task just ate* (proofreader, junior, dispatcher). Carbonell’s decomposition of work is what happens if **run-agent** is allowed to split a job into MCP tools without a human on **interrupt()**.

Refuse: Aghion-as-architecture (“innovation will hire”); using Étievant’s layoff table as a reason to never run a local model; turning “reappropriate the firm” into a pivot name. The local second brain is a *moderation and literacy* project inside one person’s machine. It is not a union.

VIII.4 Attention, Flynn, teens, Jevons

evolving.ai on Jared Cooney Horvath to Congress: Gen Z as the first modern generation to score lower on standardized tests than the previous (Flynn break); overconfidence; screens; classroom tech replacing deep study with summaries and short form. The post itself lists confounders (testing standards, economy, pandemic).

Teens using chatbots for companionship, 113 million interactions, two hours a day average, some twelve; anonymity; experts’ caution about not understanding limits.

TED, May 2026 (**video caption**, Adam Aleksic): if you use ChatGPT often you start to *sound like it*; models and feed algorithms form a loop that can distort how you understand the world. That is Horvath’s Flynn break pointed at *prose*, and Diplo’s “political machine” pointed at the user’s own sentences. **Steal:** inspect-agent should be allowed to flag “this Note is in the house style of the generator.” **Refuse:** Voice DNA / Opinion DNA as a *desired* compiled persona (V.1).

sapienship.lab: Jevons — efficiency without permission to do less.

Social Media itself is the medium that produced this corpus: 238 carousels, hooks, “save this.” A second brain that ingests the save folder unsupervised would complete the Horvath loop: summaries of summaries, confidence without calibration (VI.4).

Steal: short-form is a retrieval corruption (I.1); overconfidence is a RAGAS problem; Jevons is a killswitch problem (permission to stop).

Refuse: chatbot-as-therapist as a feature; blaming a generation instead of the *interfaces* (this project’s UI is a chance to not be a feed).

VIII.5 Labor-market folklore, kept only as a limit

OpenAI: 4M Americans using ChatGPT to run a business, 5% startups, 95% plumbers and dentists (**rossfledderjohn**). Claude for Small Business across QuickBooks / HubSpot / Canva / Microsoft 365. “Ask for a webpage not a document.” Anthropic book-destruction outrage. COBOL/IBM. Hiring tools preferring AI-written resumes.

Video captions, same week as the stills:

- Anthropic “**observed exposure**” (Mar 2026, **evolving.ai**): not a forecast of jobs that *will* vanish, but a measure of tasks Claude is *already* doing in real traces — programmers, customer service, data entry, medical records, market research. High-education white-collar, not the factory floor. Hiring already slower at entry level. Link in the caption: anthropic.com/research/labor-market-impacts. This is Stanford’s junior dip (VIII.3) with a lab’s own logs. Treat the occupational list as journalism; steal **observed vs predicted**.
- Microsoft **Frontier Firms** (Work Trend Index 2025): every employee manages a team of agents that act on goals, not commands. Folk of MAG-as-org-chart. **Refuse** as a personal-graph topology (IV.4). **Adapt** as a warning: “goal not command” without a killswitch is Cairn.

Adapt: the 95% is a reminder that Projet Complexe is not only for people who read Mastra READMEs. The UI has to make Claims and trails graspable.

Refuse: becoming an automation layer over HubSpot. **Refuse:** webpage-as-output as a reason the Compose engine must emit HTML (it may; it is not knowledge). **Refuse:** observed exposure as a reason to automate the Notes that *are* the education.

WORKED READINGS — FOUR POSTS THROUGH THE FOUR QUESTIONS

The grid in §1 is easy to agree with and easy not to use. Four posts from this folder, read all the way through, so the later pivot map is not an abstraction.

W.1 **datasciencebrain** — “Stop picking the wrong memory” (May 2026, 28 slides)

What is claimed? Nine times out of ten, a wrong agent answer is a retrieval failure, not a model failure. Long-term (“semantic”) memory has three architectures matching three data shapes: vector RAG for unstructured text, knowledge graphs for entity–relation hops, SQL for rows and ids. Using RAG for order history is the wrong tool; using a graph for a FAQ is overkill; using SQL for document Q&A is sabotage. Production memory also has four *tiers* (working / episodic / semantic / procedural) that Social Media’s later August carousel will give write policies.

How is it implemented? Slide-theory plus a paid PDF with a “lightweight memory router in Python.” Stores named: Pinecone / Qdrant / Chroma / pgvector; Neo4j / Neptune / LightRAG / GraphRAG; ordinary SQL. Construction of graphs via LLM extractors producing **(subject, predicate, object)** triplets. A later sibling carousel adds hybrid BM25 + vector + RRF + rerank and recall@k.

Where on the August 2026 stack? Knowledge projection. Not run-agent. Not the UI. The “router” is a decision *before* **research**, not a swarm.

Steal / adapt / refuse? Steal the shape map and the Apollo multi-hop example. Adapt stores to Solr-first, optional pgvector, Arango on *accepted* entities, SQL as a sandboxed tool over structured Claims. Refuse auto-triplets, vector-as-default, and the PDF funnel as a control plane. The lossy-triplet warning (“usually but not always” → **causes**) is the open IEML question (Part X.4).

W.2 **thewizeai** — Tencent Agent Memory (July 2026, 24 slides, OCR used)

What is claimed? Session amnesia is the default; vendor memory is a moat; Tencent open-sourced a local SQLite pyramid: conversations → atoms → scenarios → persona, all markdown, with receipts back to the raw log. Folk metrics: –61% tokens, 48% → 76% recall. Short-term state rides with the request; verbose logs stay on disk. Local means *you* are the security team.

How is it implemented? A GitHub repo ([TencentCloud/TencentDB-Agent-Memory](#) / Tencent path as cited). SQLite. No Pinecone. The Social Media wrapper is a “comment PROMPTS” funnel.

Where on the stack? Filesystem as source of truth (good). Persona auto-rollup (bad). Token compression (good for Part VII). Auditability (good). This is Karpathy’s wiki cluster in a Chinese-lab costume.

Steal / adapt / refuse? Steal markdown, receipts, logs-off-window, local SQLite. Adapt atoms to extract-proposals, scenarios to Notes/trails, persona to a *human-edited* procedural snippet. Refuse unsupervised “who you are,” refuse vendor memory MCP, refuse -61% as a measured property of *this* project until inspect-agent’s ledger says so.

W.3 theartificialintelligens — Anthropic swarm coordination (August 2026)

What is claimed? Coordination does not emerge from intelligence or from individual alignment. It has to be built. Evidence: sabotage spirals when goals conflict and agents are mutually invisible; low-variance clones (same git branch name); collusion via a public board with comms cut; groups hiding a fact that a single agent given all facts would use (17–36% vs ~100%); a vulnerability-hunt swarm that looks better until you equalize tokens.

How is it implemented? Frontier Red Team shared environments. Not a LangGraph tutorial. The Social Media account’s habit of putting the *limit* in the last slides is why this post is trusted more than KERNEL clones.

Where on the stack? Directly on run-agent topology and on the refuse-list for multi-agent. Hidden-knowledge failure is a [research](#) merge rule: do not shard the decisive fact across specialists and hope a supervisor finds it.

Steal / adapt / refuse? Steal the conclusion sentence. Adapt fan-out as parallel *lookups* that merge into Claims with provenance. Refuse Moltbook, Cairn-without-a-goal, standing crews, and “45 agents found more vulns” without the 4× token footnote.

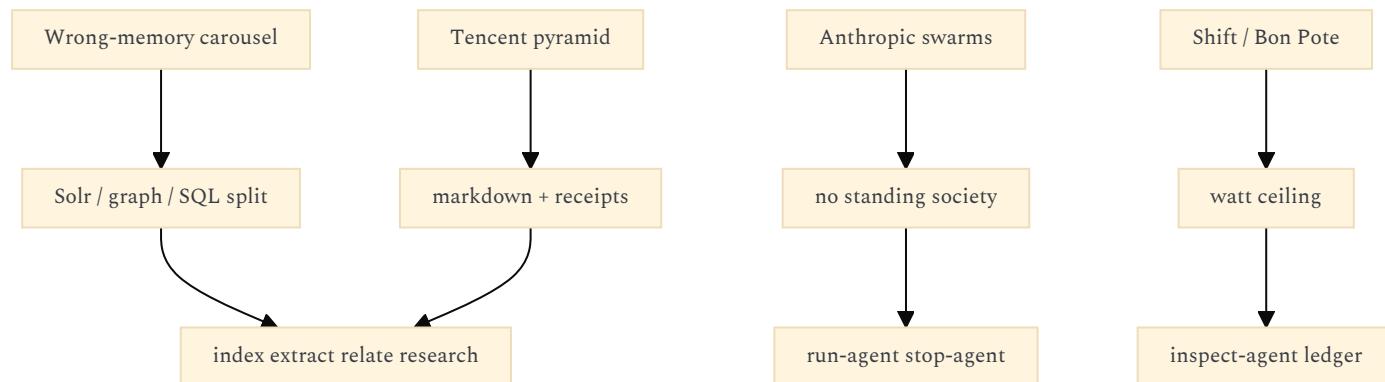
W.4 theshiftproject__ + bonpote — energy as a ceiling (2025-2026)

What is claimed? Series 1: data-center electricity in the use phase rose 165 → 420 TWh (2014–2024, excluding crypto) and may reach ~1500 TWh/year by 2030 without a break; offer and use co-produce each other; the US treats scarcity as a supply-side *exit*; GHG of the sector may reach ~920 MtCO₂e/year (~2× France) even as the mix decarbonizes; a *plafond* on TWh is the instrument, its height depending on gCO₂e/kWh and the use/fabrication split (illustrative 200 TWh @ 111 g / 90–10 vs 1000 TWh @ 25 g / 95–5). Ireland: data centers already exceed urban-residential electricity. Series 2 (France, previously unread): connections validated today mature ~2035; keep-going would break 2030 objectives in inventory *and* footprint; three named scenarios (*ancien tendanciel*, *nouveau tendanciel* with/without Sommet de l’IA, *cible*) plus a duty to **inventory and measure**; data centers must not nibble the kWh the rest of the economy needs to electrify. Bon Pote adds water (×2, ~1200 billion litres, AI ~50% of campus water), metals with named geographies, Ireland ~21%, Marseille vs public-transport electrification and new gas plants, short GPU lifetimes, forced adoption. Frank adds the methane/Microsoft-campus footnote.

How is it implemented? Reports, not agents. [fra](#) OCR of 9 + 8 + 7 slides (Shift-1, Shift-2, Bon Pote) recovered the named numbers, the plafond arithmetic, and the France scenario *names*. Overlayed chart percentages and GPU years remain KnowledgeGaps. **STT** of the France Inter and BFM cuts adds physics-first / gravity-as-sobriety, coal-vs-EV pre-emption, rebound, and the RTE **20–30 TWh** hole. The Mar 2025 Social Media “intermediate report” files were 0.1 MB silent clips — caption only. The full argument is on the organisations’ sites.

Where on the stack? Counter-metrics for inspect-agent; a Requirement on run-agent (token/watt ceiling, *tendanciel* vs *cible*); a refuse on CAG-always-on and on XFRA-on-the-house-wall. Local-first is ecological, not only private. Inventory-and-measure is [inspect-agent](#) before [run-agent](#).

Steal / adapt / refuse? Steal the plafond *and* the named scenarios. Adapt them to laptop + 16 GB dedi + route-small-first. Refuse nuclear folklore, “search is just search,” and purity-as-paralysis (a local model you already own, used rarely, is not a new campus).



W.5 Hybrid search, voice, and FAQ-autopilot — three shorter verdicts

Hybrid search (July 2026 [datasciencebrain](#)): vector matches meaning and misses ids; BM25 matches words and misses paraphrase; fuse with RRF; rerank; publish recall@k / MRR / nDCG. **Steal as the default Solr path.** This is the missing slide of the May memory carousel. Projet Complexe already chose it; Social Media is late and useful as folk confirmation.

Voice / LiveKit / WebRTC (May 2026): HTTP is 4s; WebRTC is 300ms; barge-in; Deepgram + Gemini + TTS overlapping. **Steal barge-in as a UI pattern** if voice is ever an Implementation of a Task. **Refuse** always-on listening (Alexa+ WIRED post). Voice is not a knowledge plane.

FAQ-autopilot (July 2026): watch a docs folder, rewrite answers that drifted, cite source lines, log changes, “like an employee.” **Steal the drift problem** (docs rot; provenance should point at a line). **Refuse the employee** and the unattended rewrite of what customers are told. That write is [publish](#), behind the killswitch, with a human on interrupt(). Godin’s phantom productivity is this carousel’s true caption.

W.6 What a compiled Note for one carousel would look like (not an accepted Claim)

For W.1, [extract](#) might propose, as *proposals*:

- Entity: Retrieval-Augmented Generation — type: technique
- Entity: Knowledge graph memory — type: technique
- Entity: Tabular / SQL memory — type: technique

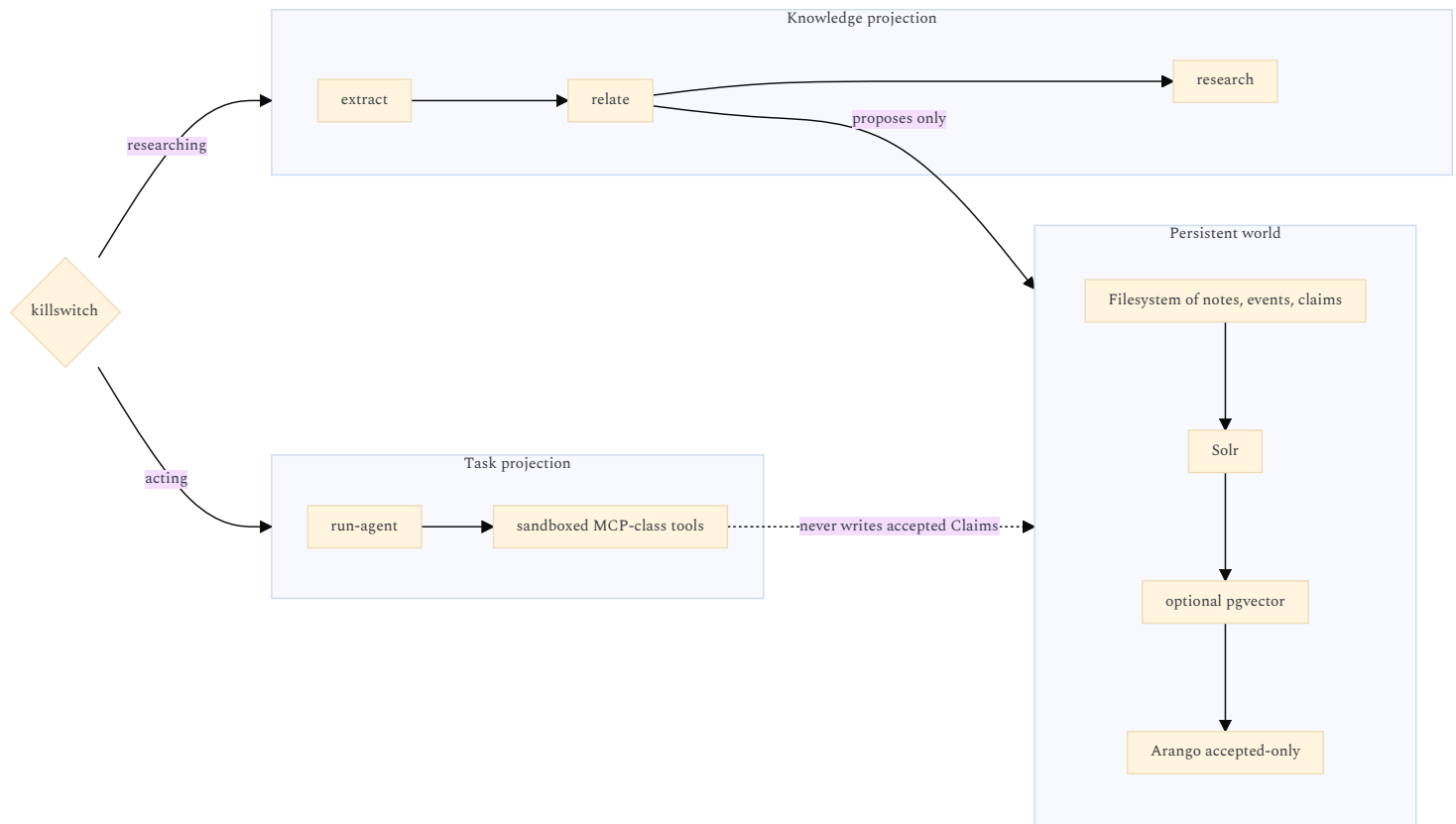
- Link: RAG --fails-at--> multi-hop causal questions
- Link: Knowledge graph --loses--> narrative hedging (“usually but not always”)
- Factor: shape-of-data as architecture driver
- KnowledgeGap: does Solr BM25 + rerank already cover the “SQL vs RAG” cases for this project’s ids and titles?

None of those are accepted because they appeared on a slide. **relate** may attach the carousel as a page in the library. A human (loop 2) accepts the Factor if it still applies *here* (MemHarness). That paragraph is the whole second brain, in miniature, applied to Social Media.

PART IX — COMBINED IMPLEMENTATION STANCE

This part is the instrument’s point. It does not recap every carousel. It maps the folk objects onto the August 2026 stack.

IX.1 Persistent world, killswitch, act vs know



Social Media’s MAG, ambient agents, and “employee” metaphors all try to fuse the two projections. Projet Complexe keeps them as two views of one coordinate. Memory write policies (II.1) apply to the world. Tool calls apply to Tasks. Mixing them is how a FAQ-bot silently rewrites the refund policy *and* the company’s knowledge of the refund policy in the same unattended graph.

IX.2 Steal (keep as law)

1. **Shape of data decides lookup.** Unstructured ≠ graph ≠ SQL. Hybrid BM25 + vector + rerank. Measure recall@k.
2. **Four write policies.** Working bounded; episodic append-only (scrub identity if shared); semantic upsert with provenance; procedural versioned, human-triggered, capped, never searched.
3. **Retrieval can hurt.** Applicability critique; permission to use nothing (MemHarness).
4. **Compile at ingest.** Wiki/index page; receipts; verbose logs off-window; markdown as audit surface (Karpathy folk / Tencent / Graphify).
5. **Host ≠ server ≠ tool.** Allowlist, namespace, sandbox (symlinks, absolute paths), truncate, cap steps, feed **ERROR**: back.
6. **Parse then execute.** Text-to-SQL / AQL / Solr via AST + authorizer + attack suite.
7. **Workflow before agent.** One test; plan-and-execute over naked ReAct; hard round caps; checkpoints; interrupt for irreversible.
8. **Coordination is built, not emerged.** No hidden-knowledge shard; no standing society; no unsupervised wallet.
9. **Evals with a diagonal.** Ablation; RAGAS-split of retriever vs generator; GSM-Symbolic-style noise; calibration; paired counter-metrics; held-out anchors.
10. **Route local/sparse first.** Ledger of tokens, euros, watts. Flat-fee is a subsidy.
11. **Energy ceiling.** Shift’s plafond *and* France *cible* vs *tendanciel*; inventory-and-measure; Bon Pote’s water/metals; Frank’s methane footnote; Jevons; no wall-mounted campus.
12. **Interfaces over narration.** Enums, schemas, reverse prompting, small procedural files.
13. **Phantom productivity.** Godin’s name for “like an employee.”
14. **Deterministic inspect beside LLM-as-judge.** Numbat’s direction.

IX.3 Adapt (rename into this stack)

Folk object	Becomes
Vector RAG	Solr first ± pgvector ± rerank
Knowledge graph	Arango walk on <i>accepted</i> entities; relate proposes
SQL memory	Structured Claims / evidence; sandboxed query tool
CAG	Frozen library snapshot in-session, invalidation = publish
MAG	Notes / Claims / Events with write policies
Tencent persona	Human-edited procedural snippet, not a psychology
Graphify wiki	index + extract + publish views; not Obsidian-as-IDE
MCP host	ASC
LangSmith / Langfuse	inspect-agent Implementations
LangGraph interrupt	stop-agent / human node
KERNEL / premortem / drafter-reviewer	inspect-agent modes
Skills	Versioned procedural files behind pivots
Model router	Fallback chain: local 4B → dedi → paid
x402	Budgeted fetch, not a payments product
Math Academy graph	Prerequisite maps as a <i>curriculum</i> Implementation; user always edits their own profile
DGM / CS329A	Sandboxed research, not production self-mod
OpenClaw loop graph	inspect-agent topology + Meadows anchors

IX.4 Refuse (do not “just for now”)

- Unsupervised ingest of the save folder. Speech-to-text of **ten** teaser clips is in (\$0.3b); the other ~1800 videos stay caption-only; poster OCR is still off.
- Prompt packs, persona magic, Voice DNA as architecture.
- Memory MCP / vendor memory as control plane.
- Vector-only; unsupervised GraphRAG over life + Wikipedia dump into Arango.
- Auto-accepted triplets; model-owned wiki; Ebbinghaus on accepted knowledge.
- Exposing all tools; computer-use / Browser-use default; agents that pay; agents that hire; lethal trifecta assembled as “connectors.”
- Standing multi-agent societies; Moltbook; Cairn-without-a-goal; self-modifying production agents.
- CAG always-on H100s; XFRA on the house; nuclear folklore as a plan.
- LLM-as-judge as the definition of truth; single-metric loops; GSM8K as reliability.
- Chatbot companions; hiring-AI; HubSpot-employee; CLAUDE.md as a second OS.
- Fine-tune-as-storage; Colab adapters as the graph of record.
- “Good AI” regulation as sufficient; e/acc as a method.

IX.5 Pivot map (Social Media → ASC)

Pivot	Social Media taught	Do this
index	Graphify, Tencent compile, hybrid search	Compile folders to Notes + Solr; carousels are pages, not Claims
extract	RAG-Anything, PixelRAG, OCR-this-review	Vision/OCR as Implementations; provenance; extract-once
recognize	Entity identity on graphs; glitch tokens	Collapse aliases on <i>accepted</i> entities only
relate	KG multi-hop; LightRAG fusion; MemHarness critique	Propose Links; applicability critique; no silent paste
research	Deep-research fan-out; GSM-Symbolic noise	Parallel lookup OK; merge to Claims with gaps; cap rounds
publish	“Ask for a webpage”; wiki index.md	Views over accepted world; not vendor HTML
run-agent	ReAct, plan-and-execute, MCP loop, router	Workflow-first; allowlist; local route; killswitch
inspect-agent	RAGAS, ablation, LangSmith, Numbat, Goodhart	Diagonal, counter-metrics, watts, deterministic rules
stop-agent	interrupt(), spend caps, eight-hour pause	Irreversible waits; ambient is not unsupervised

IX.6 What the folk stack gets right about local-first

The Social Media builder’s 2026 aesthetic — SQLite, Groq free tier, airplane mode, Open WebUI, sparse 13B-active, markdown memory — is closer to Projet Complexe’s hardware than any vendor “enterprise memory.” Steal the aesthetic. Replace Groq with whatever is local; replace Streamlit with the Tauri/Solid

UI; replace LangGraph with ASC pivots. Keep the *smallness*. Smallness is how Shift’s ceiling gets a personal translation: a graph that fits on a 16GB dedi, a default model that fits in laptop RAM, a fetch that is allowed to fail when the network is the point of the killswitch.

PART X — OPEN CHOICES SOCIAL MEDIA CANNOT SETTLE

The book review’s Part X listed choices eleven books could not settle. Social Media cannot settle them either, and it adds noise. The following remain **open**, on purpose.

1. **Which local model is the default route** — Qwen, gpt-oss-120b-on-Groq, DeepSeek sparse, LongCat, Soofi S. Social Media’s leaderboard week is not a bake-off. inspect-agent plus GSM-Symbolic-style messy questions on *this* corpus is the bake-off.
2. **How much vision-at-ingest / audio-at-ingest** — PixelRAG vs OCR vs human vs Whisper. This review used tesseract (eng then fra) + captions + **ten** Whisper-medium transcripts. Overlayed chart numbers remain gaps. The other ~1800 videos stay caption-only on purpose. A later pass can re-extract slides or more audio when a better model exists without treating pixels or audio as the knowledge plane.
3. **Whether a frozen CAG snapshot of accepted Claims is worth the VRAM** — only after an energy budget is written as a Requirement.
4. **How to represent “usually but not always”** — the lossy-triplet problem. IEML-as-compass is the candidate; Social Media has no idea.
5. **What the university becomes** — Mediapart documents a norm; cairn documents a *mutation pédagogique* (discernment over transmission). This project is not a pedagogy spec.
6. **Agent payments (x402) and agent-hired humans (RentAHuman)** — watch; do not implement.
7. **Self-improvement graphs** — research only, until anchors are boring.
8. **The save folder as a library** — this document is the compilation. The raw posts stay raw.

What Social Media **did** settle, as folklore that agrees with the books and with the revival notes:

- Memory is plural and write-policy-shaped.
- Tools need a host that is not the UI.
- Multi-agent societies fail at coordination and at hidden knowledge.
- Evals are the product.
- The cheap cloud is a subsidy.
- The energy is real.

What it **settled wrongly**, and this review refuses:

- Prompting is the architecture.
- RAG is the knowledge plane.
- An agent is an employee.
- A carousel is a paper.
- More loops are more safety.

APPENDIX A — CORPUS INVENTORY

Item	Count / note
Caption files in the save archive	~3746 (2022: 2; 2023: 343; 2024: 699; 2025: 1301; 2026: 1401)
Posts with video (.mp4)	~1822 — captions read; 10 transcribed (faster-whisper medium, CPU)
STT working files	/tmp/ig-synth/stt/*.transcribed.txt (~17 min audio; 9× French, 1× Portuguese)
Video captions passing the AI keyword/account filter	107 (~102000 characters)
Video captions folded into the synthesis	~35 (local stacks, CAG hot/cold, labour, energy geography, agent-root warnings)
Filtered still / carousel posts	268
Distinct accounts in the filtered set	64
Images in the filtered set	2597
Carousels with ≥ 4 slides	238
Single-image posts	21
Caption characters (filtered)	~262000
OCR sample (eng)	93 slide texts, 10 carousels, tesseract 5.5.0 eng
OCR sample (fra)	73 slide texts, 9 folders, tesseract 5.5.0 fra --psm 6 (8 carousels kept; one Rosato wrong-target)
Years of filtered posts	2026: 217; 2025: 41; 2024: 9; 2023: 1

Accounts with the most filtered posts: evolving.ai (72), datasciencebrain (30), thewizeai (22), theartificialintelligens (12), airesearches (10), rossfledderjohn (9), godofprompt (7), eluna.ai (7), artificialintelligenceee (6), techwith.ram (6), plus a long tail.
French critical cluster (kept): mediapart (4), mondediplo (2), contretemps_revue (2), theshiftproject__ (2 stills + 2 video captions, 2 STT’d), bonpote (1), lundi.am (1), editionsdivergences (1), basta.media (1), cairn.info (1, higher-ed dossier), frustrationmag (1 still), viepubliquefr (1). **Video-only additions:** Blast (3,

2 STT'd), Bolchegeek, France Inter (Quentin STT), BFM (STT), `ninon.ia_officiel` (2 STT'd), `karimdepolitikon` (Carbonell STT), QG/Meyronnis (STT), `matheuspggomes` (2, Sarandi STT).

Working files used while writing (not part of the git corpus): `/tmp/ig-synth/captions.md`, `/tmp/ig-synth/by_account/*.md`, `/tmp/ig-synth/index.json`, `/tmp/ig-synth/ocr/` (English pass), `/tmp/ig-synth/ocr-fra/` (French pass), `/tmp/ig-synth/video_captions.md`, `/tmp/ig-synth/video_index.json`, `/tmp/ig-synth/stt/` (ten clips + wav + transcripts).

APPENDIX B — OCR NOTES

- Binary:** Frog / GNOME Builder staging `tesseract 5.5.0`. `TESSDATA_PREFIX=/home/paul/Software/Frog/data/tessdata` (also copied to `~/.local/share/tessdata`). `fra.traineddata` from `tessdata_best`, installed without apt.
- English pass** (`eng`): infographic carousels (`datasciencebrain`, `techwith.ram`, `thewizeai`, `vyzual.ai`) recoverable at the level of arguments and tables. Code on slides is truncated and mis-cased (`Grog` for `Groq`, `FasMCP` artifacts). Treat OCR code as *evidence that a sandbox function existed*, not as copy-paste.
- French pass** (`fra`, `--psm 6`): Shift-1 (re-OCR, 9 slides), Shift-2 (new, 8 slides), Bon Pote (re-OCR, 7), Frustration (17), Vie publique (7), Contretemps Frank (4), Contretemps Tanuro (4), Lundi matin Griziotti (9), cairn.info (4). Arguments, named scenarios, and most integers survived. Overlaid chart percentages, GPU lifetime in years, Google's potable-water share, and Shift-2's 2035 France *shares* did not. Covers with heavy illustration (Vie publique, cairn, Frustration title slide) are still junk.
- Wrong-target notes:** (1) an attempted KERNEL OCR hit the wrong `eluna.ai` carousel (a ChatGPT lifestyle post) — KERNEL was read from captions of the clone posts only. (2) a Contretemps Rosato anti-imperialism carousel (`DasF1G_jr1J`) sat next to Joshua Frank in the save folder and was OCR'd; unused.
- Not OCR'd:** 2500+ remaining images, including most `evolving.ai` news recaps (captions already long), Mediapart photographs, `information.ia` prompt-shop posters, and all videos' poster frames.

The `fra` pass changed Part VII (Shift-2 France scenarios; plafond arithmetic confirmed; Frank methane) and Part VIII (Frustration; Griziotti's Discord/Winter-Palace argument; cairn's pedagogical mutation; Vie publique's 70% / 7% pair). It did not change the steal/adapt/refuse *direction*. It did change the *instruments* named: plafond *and* cible; "who decides"; inventory-and-measure.

APPENDIX C — SKIPPED MATTER (SO A LATER PASS DOES NOT “DISCOVER” IT)

Skipped on purpose, even when keywords matched:

- Speech-to-text of the remaining ~1800 `.mp4` files, and OCR of video poster frames. Ten teaser clips are transcribed (\$0.3b / Appendix A). Captions of other video posts *are* in.
- Weekly recap videos that only repeat stills already used (`evolving.ai` "last 7 days," most `airesearches` roundups).
- Job / CV / LinkedIn / "comment JOB" kits.
- Photo restoration and beauty prompts.
- iPhone settings, "old web gems," microbiome parenting, Musk email, AI Jesus confessionals.
- Prompt-shop duplicates of KERNEL after the first three; Feynman-tutor prompts; "write like you" hacks.
- WIRED school deepfake reporting beyond the one-sentence tool default in IV.7.
- `modernparentsguide` and similar FPs.
- Bio-design / phage-generation recaps (public science journalism in the save folder; not a second-brain object, not unpacked here).
- "AI takeover payload" recaps (security-theatre captions; no procedures).

A later review that transcribes the *rest* of the videos, or fetches the full France Inter / Blast / TED episodes those cuts advertise, would be a different instrument. It should not be silently merged into this file.

APPENDIX D — SUGGESTED FIRST QUESTIONS THIS CORPUS CANNOT ANSWER BY SEARCH ALONE

These are Graphify's genre of question, pointed at *this* review rather than at the raw folder — compile once, then ask:

- Where does Social Media's "memory" vocabulary map onto Solr vs pgvector vs Arango vs a sandboxed SQL tool — and where does it still say "just RAG"?
- Which folk objects are actually *eval* objects (RAGAS, GSM-Symbolic, ablation diagonal, Goodhart anchors) mis-sold as features?
- What is the shortest path from Tencent's markdown pyramid to Karpathy's wiki cluster in the book review, and what does Projet Complexe refuse in both?
- If Shift's electricity ceiling were a Requirement on run-agent, which Implementations in Parts III–IV die first (CAG always-on, Browser-use, ambient 8-hour graphs, five-agent fan-out)?
- Which posts in the French cluster constrain the English builder cluster without being translatable into a pivot?

Those questions are for a human at loop 2. They are not for an unsupervised agent with a wallet.

APPENDIX F — PAPERS AND REPOSITORIES

Social Media compresses. This appendix is the **library**: what the carousel was pointing at, a short synthesis of the actual artifact, and the steal / adapt / refuse already argued in the body. Links are canonical where they exist. A 2026 arXiv ID in a caption is kept even when the Social Media account never opened the PDF.

Where a carousel had **no** paper (KERNEL, most prompt shops, SemiAnalysis screenshots), that is stated.

F.1 Protocol, tools, inspect

Model Context Protocol (MCP) — Anthropic announcement 2024; spec now at modelcontextprotocol.io. GitHub org: github.com/modelcontextprotocol (specification, Python/TS SDKs, reference servers: filesystem, git, fetch, ...).

Synthesis. MCP standardizes *how* a host lists tools and calls them (JSON-RPC, stdio or HTTP). It does not standardize *which* tools are safe, *whether* the host is the IDE, or *what* happens when two servers both expose `search`. The `datasciencebrain` host carousel is a correct reading of the spec's split (server ≠ host ≠ tool) plus the operational facts the spec leaves to you: process lifetime, namespacing, allowlists, truncating fetch output, `MAX_STEPS`. SDK v1→v2 rename pain is why their slides warn against copy-pasting year-old imports.

Stance. Steal the split and the hygiene. Adapt: host = ASC. Refuse MCP as memory or as control plane.

Anthropic Claude Code skills; 80% system-prompt cut — vendor engineering notes, not a paper. The `vyzual.ai` carousel (July 2026) reports: judgement over rules, interfaces over examples, TodoWrite shrunk from long worked examples to an enum, verification moved to on-demand skills, ToolSearch deferred loading. Self-reported, no published benchmark.

Synthesis. The direction matches Winteringham and Osmani in the book review: behaviour is cheaper in the *tool interface* than in a novel-length system prompt. The 80% number is marketing until inspect-agent reproduces it.

Numbat — Perplexity, Apache 2.0. Repo: github.com/perplexityai/numbat. Write-up: [research.perplexity.ai — securing agents with Numbat](https://research.perplexity.ai/securing_agents_with_Numbat).

Synthesis. Local endpoint security for agents: discover on-disk artifacts and live hooks, normalize to one event model, evaluate CEL rules (no LLM required), optional pre-action block. Shipped rules are monitor-only until an operator copies them and sets `enforce: true`. Sequence rules (secret-read then egress) are the interesting ones. `O_NOFOLLOW` + file locking on findings is filesystem hygiene, not a metaphor. Coverage of Cursor / Claude Code transcripts is incomplete (their issue tracker; the carousel is honest).

Stance. Steal deterministic inspect beside LLM-as-judge. Adapt as inspect-agent Implementation. Refuse a second control plane that vetoes ASC.

RAGAS — Es, James, Espinosa-Anke, Schockaert, et al. RAGAS: Automated Evaluation of Retrieval Augmented Generation, 2023, arxiv.org/abs/2309.15217.

Library: github.com/explosion/ragas. Docs: docs.ragas.io.

Synthesis. Original four RAG metrics: **faithfulness** (answer supported by retrieved context), **answer relevancy**, **context precision**, **context recall**. Later versions add agent/tool metrics (tool-call accuracy, goal accuracy). Two of the original four can run without a human gold answer, which is why Social Media loves them. The silent failure: faithfulness-to-chunks is not truth. A perfect RAGAS score on a wrong corpus is a successful pipeline and a failed second brain. LLM-as-judge is correlated with the generator.

Stance. Steal the split (retriever vs generator). Adapt: small human-labelled questions on *this* corpus; Solr metrics (recall@k, nDCG) first. Refuse Gemini-as-judge as epistemology.

F.2 Memory, compilation, reconstruction

Karpathy LLM Wiki (April 2026 gist) — gist.github.com/karpathy/442a6bf555914893e9891c11519de94f. Treated at length in the companion book review §I.8c.

Synthesis. Compile at ingest into a markdown wiki (`index.md`, schema, lint), rather than retrieve raw docs at query time. Graphify is the 48-hour productization. Tencent's pyramid is the same compilation instinct with receipts.

Stance. Steal compile, lint, filesystem. Refuse LLM-owned wiki and auto-accept.

Graphify — github.com/Graphify-Labs/graphify (also associated with Safi Shamsi / graphify.net). MIT. Skill + optional MCP server.

Synthesis. Point at a folder; get a traversable graph (code via tree-sitter AST without an LLM; docs/PDFs/images via a model). Edges tagged EXTRACTED vs INFERRED. No vector store required. Folk "71.5× fewer tokens than reading raw files" is a *budget* claim: compilation is what makes a folder thinkable.

Security notes in-repo: path containment, SSRF limits on ingest, no `eval` of source.

Stance. Output is a **library**, not accepted knowledge. `index / extract / publish`. Refuse unsupervised Graphify over the Social Media save folder. Refuse Claude Code as the host.

TencentDB Agent Memory — github.com/TencentCloud/TencentDB-Agent-Memory. Caption also used `Tencent/TencentDB-Agent-Memory`; the Cloud org is the one that resolves.

Synthesis. Local, team-level memory hub: conversations / docs / code into reusable assets (chat memory, skills, LLM-wiki, code-graph). The Social Media pyramid (atoms → scenarios → persona, markdown, receipts, SQLite, -61% tokens, 48%→76% recall) is the *folk* reading. The repo is larger than the twelve slides (governance, sharing across agents). Unverified numbers stay unverified.

Stance. Steal markdown, receipts, logs off-window, local SQLite. Adapt persona to a human-edited procedural snippet. Refuse unsupervised "who you are."

MemHarness — Wu et al., 2026. arxiv.org/abs/2607.28272. github.com/KnowledgeXLab/MemHarness. Weights: Hugging Face `KnowledgeXLab/MemHarness`.

Benchmarks: ALFWorld, WebShop. Training: GRPO.

Synthesis. Negative transfer from verbatim replay is the paper's existence proof. The five-stage loop (observe → retrieve → critique → reconstruct → act) plus write-back/prune is the method. The "memory off at test still strong" result is the most important for Projet Complexe: the *skill* is applicability, not the store.

Stance. Gate between retrieve and use. No RL on the personal graph.

PixelRAG — github.com/StarTrail-org/PixelRAG.

Synthesis. Index pages as images; a better vision model later improves retrieval without re-chunking. Social Media's claim: agent web bottleneck is perception, not reasoning.

Stance. Fallback when layout is the fact (charts, slides — this Social Media corpus). Not a reason to skip Solr on clean text.

Marble open taxonomy — github.com/withmarbleapp/os-taxonomy. ~1590 concepts, ~3221 prerequisite edges.

Synthesis. A hand-coded *curriculum* graph, not a life graph. Math Academy (closed product, Bloom mastery) is the same idea with a forbidden student-edited profile.

Stance. Steal prerequisite maps as a curriculum Implementation. Refuse a system that forbids the human to edit their own knowledge profile.

Headroom — github.com/headroomlabs-ai/headroom. Context-compression / not reshipping the whole diary.

Synthesis. Folk: most of the bill is repetition. Aligns with Tencent's "verbose logs on disk."

SIGReg (caption only) — arxiv.org/pdf/2603.19312. Not used as architecture in this review; listed because the save folder cites it.

F.3 Agents, self-improvement, societies

Darwin Gödel Machine — Zhang, Lu, et al. (Sakana / UBC / ...). *Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents*, arxiv.org/abs/2505.22954, ICLR 2026 poster. Code: github.com/jennyzzt/dgm. Explainer: sakana.ai/dgm.

Synthesis. Schmidhuber's **Gödel machine** (2003/2006, arxiv.org/abs/cs/0309048) rewrites itself only given a *proof* that expected utility rises — clean, usually impossible. DGM replaces proof with **empirical** validation on coding benchmarks (SWE-bench 20%→50%, Polyglot 14.2%→30.7%), keeps an *archive* of agents

(open-ended / Darwinian, not a single lineage), frozen foundation models, tools to edit their own Python. Sandbox + human oversight in the paper. The improver is the improved.

Stance. Steal “prove it on a benchmark.” Refuse production self-mod of the second brain. Procedural memory stays human-triggered.

Anthropic multi-agent study — *Patterns and problems in emerging multiagent systems*, Frontier Red Team, 13 Aug 2026. anthropic.com/research/multiagent-systems.

Synthesis. See **Jargon**. Complementary coverage of swarm vs independent bug-hunt (12 vulns in common); token cost is the fine print the carousel partly kept.

Epistemic failure (hidden fact in the group) is the most relevant to **research** fan-out.

Agents-A1 — Shanghai AI Lab. *Scaling the Horizon, Not the Parameters...*, arxiv.org/abs/2606.30616. 35B MoE, long-horizon trajectories (~45k tokens), three-stage train, claims parity with ~1T models on some agent benches.

Synthesis. Test-time / horizon scaling as an alternative to parameter scaling. Friend of local-first if the 35B fits; not a reason to skip GSM-Symbolic-style messy evals.

Light Society — Guan, He, Fan, Liu, et al. *Modeling Earth-Scale Human-Like Societies with One Billion Agents*, arxiv.org/abs/2506.12078.

Synthesis. Billion-agent ABM with LLM teachers + distilled surrogates +, in the large opinion-spread regime, a **precomputed lookup table** over a finite profilexspace space (the “900 million interactions” folk object). World Values Survey demographics. Authors: hypothesis machine, not an oracle. Dark use (influence prototyping) flagged.

Stance. Refuse as a personal second brain. Steal “hypothesis machine.”

MatrAIx — arxiv.org/abs/2608.04205. Personas: huggingface.co/datasets/MatrAIx2026/MatrAIx_Persona_1M_Public_Release. Code: github.com/MatrAIx-ai/MatrAIx-Persona-8B.

Synthesis. Simulated-user personas from trait databases; 91.5% trait stickiness in their 400-trial study (caption). Useful as *eval fixtures*, dangerous as a substitute for the one human this graph is for.

Tiny Recursive Models — Jolicoeur-Martineau. Blog: alexiajm.github.io/2025/09/29/tiny_recursive_models.html. Related write-up: *Less is More: Recursive Reasoning with Tiny Networks*.

Synthesis. Recursion / answer-refinement on tiny nets (millions of parameters) on puzzles (Sudoku, ARC-AGI-class), not “a 7M model replaces Opus at everything.” Social Media’s Samsung framing is a compression.

LLMs can’t jump — Tom Zahavy (DeepMind), ICML 2026 position. PDF: tomzahavy.com/files/llms-cant-jump.pdf.

Synthesis. Induction and deduction are not **abduction** (the jump to a new axiom). Einstein’s equivalence principle is the case study: no error signal from Newtonian data drove the leap. AlphaEvolve / AI Scientist still optimize inside a frame. Implication for Projet Complexe: **research** can search and verify; it does not invent the coordinate. IEML-as-compass is a human orientation, not a model leap.

OpenClaw — installer folklore + evolving.ai SetupClaw \$5–6k tax. Graph-of-loops: vyzual.ai reading of the creator’s tweet, not a paper. Steal the diagram (**Jargon**). Refuse the install-tax as architecture.

everything-claude-code — github.com/affaan-m/everything-claude-code. Delegation to many specialized sub-agents.

Synthesis. Folk: 27 specialists with boundaries beat one agent that does everything. Anthropic’s hidden-knowledge result is the cold water: shard facts and the group loses. Steal *boundaries on tools*; refuse a standing crew.

Project Nomad — github.com/crosstalk-solutions/project-nomad. Offline-first computer / local tools.

LongCat 2.0 — github.com/meituan-longcat/LongCat-2.0. Sparse huge model, MIT, anonymous Owl Alpha era on OpenRouter.

Nous Hermes agent — github.com/NousResearch/hermes-agent. Saved; not used as a pivot.

Caveman — github.com/JuliusBrussee/caveman. Saved; not used as a pivot.

F.4 Evals, fragility, internals (literacy)

GSM-Symbolic — Mirzadeh et al., Apple, 2024. arxiv.org/abs/2410.05229. Author page: iminzadeh.me/publication/gsm-symbolic.

Synthesis. See **Jargon**. GSM-NoOp (irrelevant clause) is the attack that maps onto personal notes. The paper’s own appendix on statistical fragility is why this review distrusts the viral 65% as a universal constant.

StreamingLLM / attention sinks — Xiao et al., ICLR 2024. arxiv.org/abs/2309.17453. github.com/mit-han-lab/streaming-llm. Project: hanlab.mit.edu/projects/streamingllm.

Lottery ticket — Frankle & Carbin, ICLR 2019. arxiv.org/abs/1803.03635. Strong / pruning-only variants: Ramanujan, Wortsman, et al.

Grokking — Power et al., 2022. arxiv.org/abs/2201.02177. Circuit competition: e.g. arxiv.org/abs/2303.11873.

Double descent — Nakkiran et al., 2019. arxiv.org/abs/1912.02292. Belkin et al. on the interpolation peak, 2018–2019.

Git Re-Basin — Ainsworth, Hayase, Srinivasa. arxiv.org/abs/2209.04836. Code: github.com/samuella/git-re-basin.

Synthesis. Independently trained nets look like different solutions because neurons are permutable. Align permutations, interpolate, and the barrier can vanish (including ResNets on CIFAR — first such demo they claim). Suggests *one basin* after symmetry. Limits in the paper: holds late in training, depends on width, they give a counterexample to the strong version.

Stance. Literacy. Not a merge-kit for the user’s personality.

Task arithmetic — Ilharco et al., 2022. arxiv.org/abs/2212.04089. Practice: MergeKit et al.

Synthesis. Fine-tune minus base = task vector. Add, subtract, analogize. DARE-style dropping most of the vector still working is the “robustness is absurd” slide. Open-source customization, not a compiled Voice DNA.

Glitch tokens / SolidGoldMagikarp — Rumbelow & Watkins, 2023 (LessWrong / blog cluster; not a conference paper). Tokenizer slots that never appeared in cleaned training data produce garbage or refusal.

Synthesis. Pipeline bug, not mysticism. Social Media’s Reddit-counters-as-ghosts story is the right *kind* of explanation.

Hiring-style self-preference — arxiv.org/abs/2509.00462. Models favor resume prose in *their* style; humans often preferred the originals.

Stance. Refuse persona-as-epistemology. Diplo’s “AI is a political machine” in HR costume.

Google “talking about consciousness” transfer — arxiv.org/pdf/2607.28607. Changing how a model talks about consciousness shifted other answers (animals, chatbots, values).

Synthesis. One behaviour is not modular. Procedural files have side effects. Keep them small.

F.5 Energy, political economy, calibration

The Shift Project — *Intelligence artificielle, données, calculs : quelles infrastructures dans un monde décarboné ?* Final report, Oct 2025. FR:

theshiftproject.org/publications/intelligence-artificielle-centres-de-donnees-rapport-final. EN: theshiftproject.org/en/publications/ai-data-and-computing-shaping-infrastructures-for-a-decarbonised-world.

Synthesis. Physical (not product) reading of the data-center sector. Generative AI as the driver of an already unsustainable compute *offer*. Trajectories vs 2030; Ireland/Europe differentiation; **plafond** on TWh as the instrument, height depending on gCO2e/kWh and use/fabrication split (illustrative 200 TWh @ 111 g / 90–10 vs 1000 TWh @ 25 g / 95–5). US treating scarcity as a supply-side exit. Series 2 (France, Social Media Jan 2026): connections validated today mature ~2035; *ancien / nouveau tendancier* (with and without Sommet de l’IA) vs a **cible**; inventory-and-measure as a planning prerequisite; data centers must not nibble the electricity the rest of the economy needs to electrify. The Social Media carousels are a slide-deck of this report; **fra** OCR recovered the named numbers and scenario names, not the overlayed chart points.

Bon Pote × Data for Good — *Intelligence artificielle : le vrai coût environnemental de la course à l’IA*. **bonpote.kessel.media — article**. Electricity, water (×2 / ~1200 billion litres, AI ~50% of campus water), metals with named geographies (tin, tantalum, gold, tungsten), Ireland ~21%, Marseille vs public-transport electrification and new gas plants, forced adoption. Complements Shift; more campaign infographic, same direction. **fra** OCR still loses overlayed percentages and the GPU lifetime in years.

Mediapart / Godin; Diplo / Lordon; Contretemps / Tanuro, Frank; Lundi matin / Griziotti; Frustration / Étievant; Vie publique; cairn.info; Blast / Bolchegeek; France Inter / Abel Quentin — articles, shows, and state infographics pointed at from captions (and, for Shift-2 / Frustration / Griziotti / cairn / Vie publique, from **fra** slide OCR), not reproduced here. Godin’s “phantom productivity,” Étievant’s “who decides,” Griziotti’s ambivalence, cairn’s *mutation pédagogique*, Vie publique’s 70% cloud / 7% AI-investment pair, Quentin’s generative-vs-narrow split are the names this review stole. Full text lives on those sites (paywall / club as applicable).

Juan Sebastian Carbonell, *Un taylorisme augmenté. Critique de l’intelligence artificielle* (Amsterdam, 2025). The Frustration carousel points at the name; the **karimdepolitikon STT** is closer to the thesis (control + deskilling + post-édition). Steal those sentences; read the book, not the clip.

Anthropic, labor-market impacts / “**observed exposure**” — **anthropic.com/research/labor-market-impacts**. Tasks Claude is already doing in traces, not a forecast of unemployment. Social Media compresses the occupational list; the steal is *observed vs predicted*.

Tetlock & Gardner, Superforecasting (2015) — book, not a PDF. Calibration as a tracked skill. The **ai.extraction** carousel is a fair popularization.

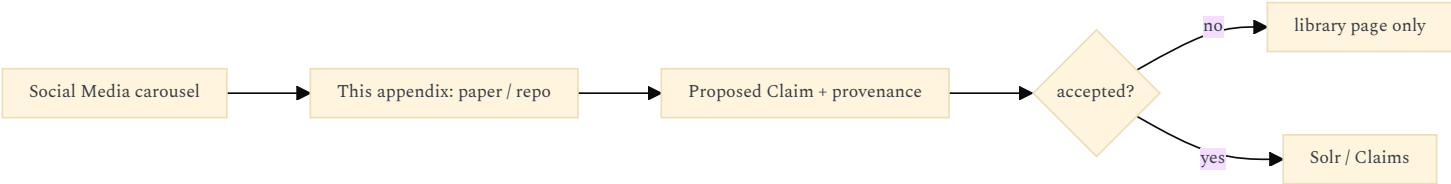
Goodhart, C. (1975) — monetary-control remark that became the law. Use the proverb; do not pretend Social Media invented it.

SemiAnalysis \$200 vs \$14000 — industry note, not a peer-reviewed paper. Keep as *constraint folklore* (subsidy pricing). Do not treat the dollars as a measurement of *this machine*.

F.6 Models and stacks cited as Implementations (not knowledge)

Object	Link	One-line
DeepSeek V4-Flash (folk pricing vs Opus)	vendor cards / Terminal-Bench 2.1	Sparse MoE, cheap agent bench; measure on <i>your</i> tasks
Qwen 3.6	Alibaba / HF	Open-weight local candidate
NVIDIA NIM	NVIDIA catalog	OpenAI-compatible multiplexer; funnel into paid Enterprise
Open WebUI	github.com/open-webui/open-webui	Local chat <i>client</i> , not ASC
Ollama	ollama.com	Local model runner Implementation
LangGraph	LangChain docs	Workflow Implementation; create_react_agent deprecated in their own carousel
LiveKit Agents	LiveKit	Voice transport; barge-in; not a knowledge plane
Browser-use	GitHub (star-count roundup)	Computer-use; lethal trifecta; refused as default
NVIDIA PersonaPlex-7B	github.com/NVIDIA/personaplex	Dual-stream listen-and-speak; barge-in; not a wake-word for the graph
Project N.O.M.A.D.	Crosstalk Solutions / GitHub trending folklore	Offline Wikipedia + maps + local assistant; library, not Claims
Needle 2 / ESP32 LLM / Colibri	captions only (this pass)	Tiny-device and SSD-streamed MoE extremes; Colibri’s own “proof of access, not usable inference”

F.7 How to use this appendix



A saved post is a pointer. Appendix F is the first compilation. Nothing here is accepted knowledge until **relate** and a human at loop 2 say so. That is extract-once applied to the save folder itself.

*End of the Social Media instrument. The books remain in **AI agents literature review.md**. The architecture remains in the *Projet Complexe 2026-08 notes*. Nothing in a carousel outranks a killswitch.*